This article has been accepted for publication in IEEE Communications Surveys & Tutorials. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/COMST.2025.3621165

A Survey on Reliability Evaluation of Optical Networks

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***Abstract*—In the past decade, rapid advancements in communication technology have fundamentally transformed methods of connection and communication in the digital age. Optical networks have emerged as indispensable communication infrastructures due to their high per-channel transmission rates, massive aggregate capacity enabled by multiplexing technologies, and excellent resistance to electromagnetic interference. The enormous transmission capacity of optical networks has greatly facilitated the development of various businesses and applications. Therefore, ensuring the reliable operation of optical networks is crucial. One of the key conditions for ensuring the reliable operation of optical networks is the ability to accurately measure or evaluate their reliability. There are many methods for the evaluation of optical network reliability. Each method has its applicable scope, advantages, and limitations. However, with the continuous development of new technologies like 6G, reliability evaluation will face higher demands and more complex scenarios. Currently, there is no comprehensive systematic literature survey on the existing optical network reliability evaluation methods. Hence, the research focuses on evaluation methods for optical network reliability. Several essential concepts in evaluating optical network reliability are clarified, factors affecting its reliability are summarized, evaluation metrics are analyzed, and evaluation methods are reviewed. Then, methods to enhance the reliability of optical networks are outlined, and the trade-offs between cost/energy consumption and reliability are discussed. Finally, some significant future challenges are described, and corresponding open research topics are proposed. This survey provides valuable references for designing more reliable optical networks or developing highly reliable operation and maintenance strategies.**

***Index Terms*—Optical networks, reliability evaluation, reliability metrics, survivability technologies.**

I. Introduction

In the past decade, services such as augmented reality (AR), virtual reality (VR), the fifth-generation (5G) mobile communications, the Internet of Things (IoT), and the metaverse have gradually boomed [1]. The popularity of these emerging services presents a considerable challenge to networks [2]. They require more online data streams to support operations. As a result, there is a higher demand for network capacity. In order to meet these heightened demands, networks must offer high-speed, large-bandwidth, and reliable transmission. Optical fiber has become the primary transmission medium in telecommunications networks due to its capacity, reliability, affordability, and scalability advantages. Optical communication technology is one of the critical technologies in the future of sixth-generation (6G) mobile communication networks [3]. It has the potential to meet high-bandwidth capacities (>1 Tbps), ultra-low latencies (microseconds), and ultra-high reliability (6 nines) in 6G networks [4]. According to the data and storage company IDC forecast report, global data was close to 100 ZB by 2022. By 2025, the total global data volume is estimated to reach 175 ZB [5]. The transmission capacity of optical networks needs to be improved to meet future high-bandwidth demands. Multiple multiplexing technologies have been applied to enhance the transmission capacity of optical networks. Representative examples include time division multiplexing (TDM), frequency division multiplexing (FDM), and space division multiplexing (SDM) [6]. In optical networks, wavelength division multiplexing (WDM), a specific form of FDM, is widely adopted [7]. WDM represents channels using wavelengths instead of frequencies. In addition, orthogonal frequency division multiplexing (OFDM) can be implemented by exploiting the orthogonality of subcarriers to improve spectral efficiency [8]. SDM increases system capacity by introducing the spatial dimension, which is typically implemented by multicore fiber (MCF) and multimode fiber (MMF), thereby multiplying the transmission capacity of optical fiber systems [9, 10]. Currently, 100 Gbps and 200 Gbps long-haul transmission systems are globally deployed on a large scale [11]. 400 Gbps transmission systems have been standardized, whereas 800 Gbps transmission systems are still in the initial stages of standardization [12].

Optical networks are playing an increasingly important role. Moreover, the reliability of optical networks is becoming equally critical. Unstable optical networks directly impact the quality of service (QoS), leading to video call delays, slow webpage loading, or even data transmission interruptions, severely affecting the user experience [13]. Furthermore, unreliable optical networks can result in severe business disruption, especially in industries like finance and healthcare, where real-time requirements are critical, potentially causing significant economic losses [14]. According to AT&T, each minute of optical fiber cutting leads to a revenue loss of approximately $3,600 [15]. Another example is Gartner Research Group, which lost approximately 500 million dollars due to optical

This work was supported in part by the National Natural Science Foundation of China (Nos. 62171050, 62125103), Fundamental Research Funds for the Central Universities (2024ZCJH02), and Open Fund of State Key Laboratory of Information Photonics and Optical Communications (BUPT, No. IPOC2021ZT15).

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network failures [16]. Therefore, ensuring the reliable operation of optical networks is crucial. However, conducting a precise reliability evaluation is a prerequisite for ensuring the reliable operation of optical networks. Without the precise reliability evaluation, it is impossible to understand the networks' actual operational state or determine their reliability [17, 18]. Optical network reliability evaluation provides a comprehensive analysis and quantitative results. Reliability evaluation ensures business continuity, fulfills service level agreements (SLAs), and guides network planning and optimization. It helps identify vulnerabilities, supports preventive maintenance, ensures SLA compliance to avoid penalties, and helps with topology design and disaster recovery technologies to enhance network robustness.

Optical network reliability evaluation is necessary and meaningful. However, evaluating optical network reliability is a complex and comprehensive process with numerous challenges. Varying deterioration rates of various components can change significantly with usage and environmental conditions. Obtaining accurate and comprehensive failure data also presents difficulties. Additionally, optical network reliability evaluation requires integrating multiple performance metrics, managing trade-offs between them, and considering the interdependencies within optical networks, which adds to the complexity of achieving an accurate evaluation.

A. Motivation

It can be seen that optical network reliability evaluation is necessary and challenging. Moreover, new technologies have further increased the importance and urgency of optical network reliability evaluation. The demand stems from three key directions for future development: capacity expansion, computing power networks, and space-air-ground integrated networks (SAGIN) [19]. While capacity expansion enhances data transmission capabilities, it also poses a risk of greater losses during failures [20]. Therefore, ensuring optical network reliability during expansion is critically important. With the proliferation of computationally intensive applications such as large-scale models, there is an increasing need to integrate and schedule computing resources in multiple different computing power centers. The distributed training of large-scale models relies on the high-bandwidth capacity of optical networks, placing higher demands on the reliability of optical networks. According to SemiAnalysis's estimate, the compute rental cost for a single GPT-4 pre-training run is about $63 million, and a single interruption can incur a loss of up to $1.1 million [21]. SAGIN combines terrestrial and satellite networks, significantly enhancing network flexibility and resource scheduling. Efficient resource management not only improves network reliability but also effectively prevents the formation of network islands under extreme conditions such as natural disasters [22]. These three development directions raise the bar for optical network reliability and impose stricter requirements on reliability evaluation methods' accuracy, scalability, and intelligence. However, no systematic review has examined optical network reliability evaluation. This survey presents a detailed analysis and summary of key factors affecting optical network reliability, evaluation metrics, evaluation methods, methods to improve optical network reliability, trade-offs between reliability and cost, reliability and energy consumption, etc.

B. Scope, Novelty, and Target Audience

Optical network reliability evaluation is a broad topic. This survey examines the factors that impact optical network reliability from five perspectives: optical components, optical cables, network topology, environment, and human factors. The reliability evaluation metrics of optical networks are outlined. The evaluation processes, advantages, limitations, and applicability of reliability evaluation methods are summarized. Survivability technologies, which can greatly enhance the reliability of optical networks in the event of failures, have always been an important research focus. Since this topic has already been extensively reviewed in the literature, including studies on survivability design of optical networks [23–26], multi-domain survivability [27, 28], survivable elastic optical networks (EONs) [29], synchronous optical network (SONET)/ synchronous digital hierarchy (SDH)-like resilience for IP networks [30], and network security against attacks [31], this survey does not provide a comprehensive survey in this work and only presents generic survivability technologies and technologies tailored to specific network scenarios. Our focus remains on the methods for reliability evaluation. Next, the trade-offs between reliability and cost and between reliability and energy consumption are discussed. Finally, the improvements in current reliability evaluation technologies, future development trends, and prospects for reliability evaluation technologies under emerging demands are summarized.

To the best of our knowledge, this is the first complete and up-to-date survey on optical network reliability evaluation. We believe that only by deeply understanding reliability evaluation can we design more reliable and robust optical networks, thereby ensuring the stability and reliability of upper-layer services such as high-definition video streaming and distributed large model training. Therefore, the target audience for this research includes students, researchers, decision-makers, industry experts, and non-professionals interested in this field.

C. Contribution and Organization

In this survey, we focus on examining the factors that influence the reliability of optical networks, the metrics and methods for optical network reliability evaluation, and strategies to enhance the reliability of optical networks. Our survey provides a valuable reference for studying the reliability evaluation of optical networks and exploring future developments in this field. Specifically, we contribute in four areas. The structure of this survey is illustrated in Fig. 1.

1) This survey introduces the basic concepts of optical network reliability evaluation. It introduces key indicators for optical network reliability evaluation and analyzes various factors that influence optical network reliability, including optical components, optical cables, network topology, environmental factors, and human factors.

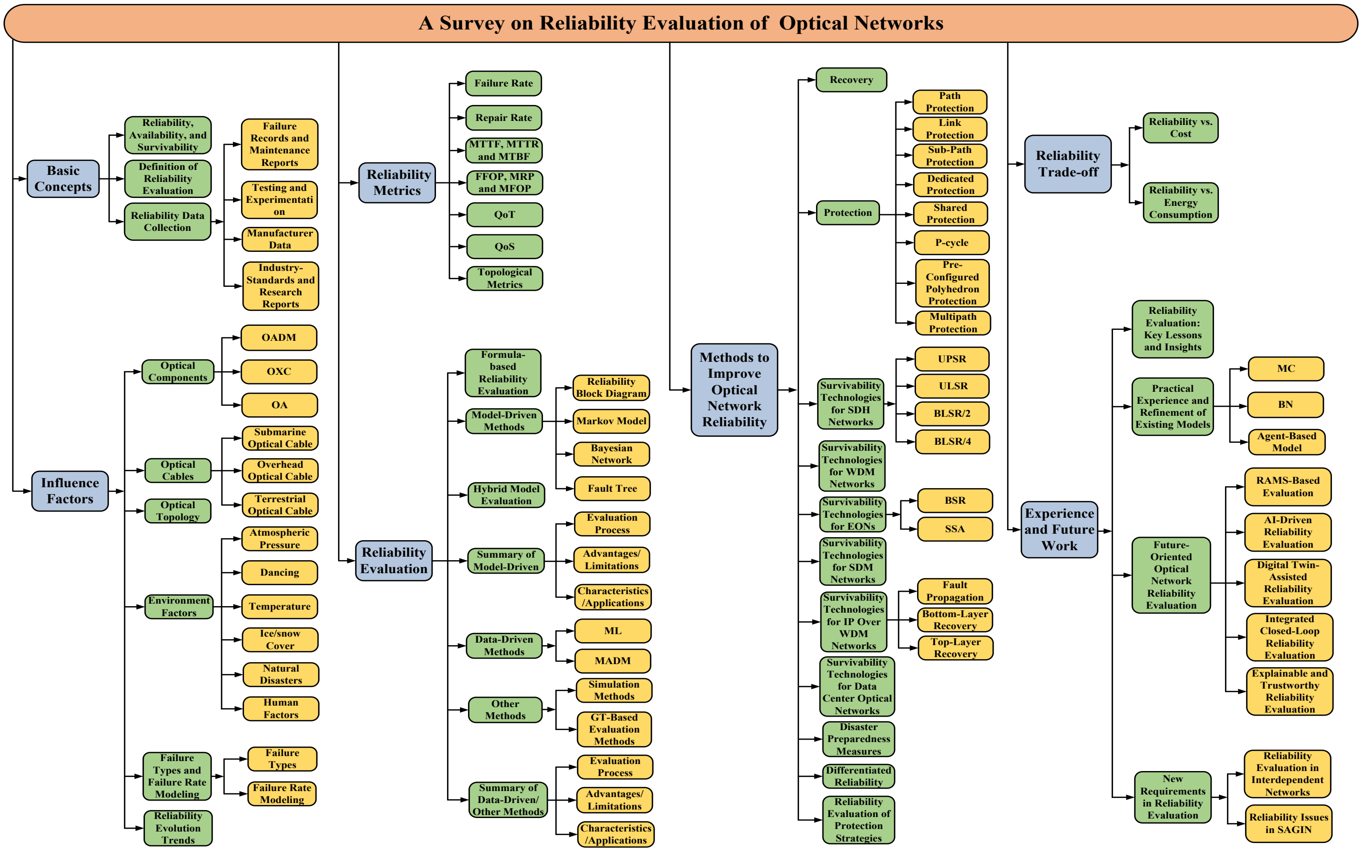

Fig. 1. The structure of this survey.

2) This survey summarizes the implementation processes, advantages, limitations, and applicable scenarios of various reliability evaluation methods, providing valuable insights for researchers in selecting appropriate evaluation technologies for different application scenarios.

3) This survey introduces methods for improving the reliability of optical networks. First, we provide an overview of general survivability technologies applicable to different types of networks. Then, we describe the specific survivability technologies unique to various networks. Subsequently, disaster recovery technologies and differentiated reliability are summarized. The reliability evaluation of protection schemes in existing research is discussed. Finally, we summarize the current hot topics related to reliability, including the trade-offs between reliability, cost, and energy consumption.

4) This survey summarizes the improvements in current reliability evaluation technologies, as well as future development trends and prospects for reliability evaluation technologies under emerging demands.

The remainder of this survey is organized as follows: Section II introduces the fundamental concepts relevant to optical network reliability evaluation. Section III examines the key factors that influence the reliability of optical networks. Section IV presents commonly used reliability metrics for optical networks. Section V reviews various reliability evaluation methods, discussing their evaluation processes, advantages, limitations, and applicable scenarios. Section VI categorizes and summarizes the approaches proposed to enhance optical network reliability. Section VII investigates the trade-offs between reliability, cost, and energy consumption. Section VIII highlights recent advancements in reliability evaluation technologies, explores future research directions, and analyzes emerging requirements in potential application scenarios. Finally, Section IX concludes the survey. A list of abbreviations used throughout this paper is provided in Table I.

II. Basic Concepts

This section first discusses the relationships among availability, reliability, and survivability, then elucidates the definition of reliability evaluation and introduces the data sources used for reliability evaluation.

A. Reliability, Availability, and Survivability

Optical networks' availability, reliability, and survivability are interrelated but differ in time scale and evaluation focus. Availability focuses on whether the network can be used now, which is the probability of operating correctly at a given instant. In practice, the target is often "five nines" (99.999%) availability, which allows no more than about 5.26 minutes of downtime per year [32]. Availability is an instantaneous metric and cannot fully reflect long-term stability [33]. Survivability assumes that failures are inevitable and focuses on whether the network can still be used after a failure. It emphasizes maintaining service continuity through survivability technologies,

TABLE I
LIST OF ABBREVIATIONS

AI	Artificial Intelligence	MILP	Mixed Integer Linear Programming
AR	Augmented Reality	MTTF	Mean Time To Failure
ASE	Amplified Spontaneous Emission	MTTR	Mean Time To Repair
APS	Automatic Protection Switching	MFOP	Maintenance-Free Operating Period
AHP	Analytic Hierarchy Process	MTBF	Mean Time Between Failure
ALD	Acceptable Level of Degradation	MEMS	Micro-Electro-Mechanical System
ATTR	Average Two-Terminal Reliability	MADM	Multi-Attribute Decision-Making
BN	Bayesian Network	NFV	Network Functions Virtualization
BSR	Bandwidth Squeezing Restoration	OA	Optical Amplifier
BER	Bit Error Rate	OCh	Optical Channel
BDD	Binary Decision Diagram	OLS	Open Line System
BLSR	Bidirectional Line-Switched Ring	OMS	Optical Multiplexing Segment
CPT	Conditional Probability Table	OXC	Optical Cross Connection
CNN	Convolutional Neural Network	OPEX	Operating Expenditure
CSPF	Constrained Shortest Path First	OSNR	Optical Signal-to-Noise Ratio
CAGR	Compound Annual Growth Rate	OPGW	Optical Fiber Composite Overhead Ground Wire
CAPEX	Capital Expenditure	OFDM	Orthogonal Frequency Division Multiplexing
DA	Double-Layer Armored	OADM	Optical Add-Drop Multiplexer
DiR	Differential Reliability	OSPF-TE	Open Shortest Path First with Traffic Engineering
DAG	Directed Acyclic Graph	PDF	Probability Density Function
DBN	Dynamic Bayesian Network	QoS	Quality of Service
DDC	Differential Delay Constraints	QoT	Quality of Transmission
DNN	Deep Neural Network	RBD	Reliability Block Diagram
EMP	Electromagnetic Pulse	RSCA	Routing, Spectrum and Core Allocation
EWM	Entropy Weight Method	RAMS	Reliability Availability Maintainability Safety
EONs	Elastic Optical Networks	ROADM	Reconfigurable Optical Add-Drop Multiplexer
EDFA	Erbium-Doped Fiber Amplifier	RSVP-TE	Resource Reservation Protocol with Traffic Engineering
FT	Fault Tree	SA	Single-Layer Armored
FS	Frequency Slot	SSA	Split Spectrum Approach
FEC	Forward Error Correction	SMF	Single-Mode Fiber
FMF	Few-Mode Fiber	SOA	Semiconductor Optical Amplifier
FDM	Frequency Division Multiplexing	SLA	Service Level Agreement
FIPP	Failure-Independent Path-Protecting	SDN	Software-Defined Networking
FFOP	Failure-Free Operating Period	SDH	Synchronous Digital Hierarchy
FPGA	Field-Programmable Gate Array	SDM	Space Division Multiplexing
GT	Graph Theory	SBN	Static Bayesian Network
GPU	Graphics Processing Unit	SBPP	Shared Backup Path Protection
GMPLS	Generalized Multi-Protocol Label Switching	SRLG	Shared Risk Link Group
HMMs	Hidden Markov Models	SHAP	Shapley Additive Explanation
IoT	Internet of Things	SAGIN	Space-Air-Ground Integrated Network
ILP	Integer Linear Programming	SONET	Synchronous Optical Network
IETF	Internet Engineering Task Force	TBF	Time Between Failure
LW	Light-Weight	TTF	Time To Failure
LWA	Light-Weight Armored	TTR	Time To Repair
LWP	Light-Weight Protection	TDM	Time Division Multiplexing
LMP	Link Management Protocol	TOPSIS	Technique for Order Preference by Similarity to an Ideal Solution
ML	Machine Learning	ULSR	Unidirectional Line-Switched Ring
MC	Markov Chain	UPSR	Unidirectional Path-Switched Ring
MRP	Maintenance Recovery Period	VR	Virtual Reality
MCS	Monte Carlo Simulation	VNF	Virtual Network Function
MCF	Multicore Fiber	WSS	Wavelength Selective Switch
MMF	Multimode Fiber	WDM	Wavelength Division Multiplexing

including protection and fast recovery. Reliability focuses on how long the system can run without failures: the probability of failure-free operation over the entire observation interval [34, 35]. It is the key indicator of long-term stability. It affects availability through the failure rate and supports survivability by providing redundant resources. Figure 2 illustrates their relationship. The intersection of reliability and survivability indicates that employing survivability technologies can accelerate the failure recovery speed of the network, thereby improving overall reliability. The intersection of reliability and availability indicates that the network can provide continuous services at the current moment (availability) and maintain failure-free operation over an extended period (reliability). The intersection of availability, reliability, and survivability indicates that the network can provide services at any instant, operate failure-free over time, and maintain or promptly restore essential services through protection and recovery technologies during failures.

B. *Definition of Reliability Evaluation*

Optical network reliability evaluation involves identifying, quantifying, and managing network states based on data and metrics. It reflects the network's ability to function under various conditions through mathematical modeling, analysis, and data processing. Key factors include component failures,

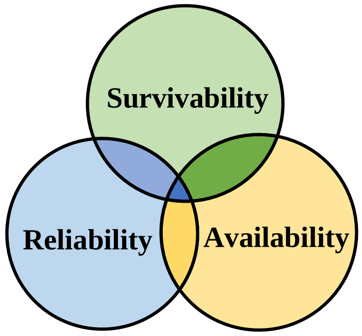

Fig. 2. The relationship among reliability, availability and survivability.

link disruptions, topology, and environmental influences. The goal is to evaluate current performance, predict future changes, and locate weak points, helping operators mitigate reliability degradation.

C. Reliability Data Collection

The prerequisite for optical network reliability evaluation is acquiring optical network reliability data, the source of which usually includes the following four aspects.

1) Failure Records and Maintenance Reports: By recording and analyzing failure events and maintenance reports of systems or components, including information on maintenance activities, maintenance cycles, etc., data on the frequency of failures, causes, and recovery times can be obtained [36].

2) Testing and Experimentation: Testing and experimentation are common methods of evaluating reliability during product development and operation [37]. Data about the reliability of the system or component can be collected by performing simulated tests under controlled conditions or observing them in a real environment.

3) Manufacturer Data: Some products or component manufacturers provide reliability data and performance specifications [38]. These data may include metrics such as the product's failure rate and design life.

4) Industry-Standards and Research Reports: Industry-standard organizations and research institutions publish reports and studies on optical network reliability, offering valuable reference data. For example, the EU COST270 project focuses on the reliability of optical components in communication systems and networks [39].

III. The Factors Affecting Optical Network Reliability

The reliability of optical networks is affected by component quality, network topology, environmental conditions, and human factors. This section provides an overview of the reliability of core elements such as optical components and cables. It then examines how topology, environmental influences, and human errors shape overall reliability. Finally, it summarizes common failure types, their failure rate modeling, and reliability evolution trends in optical components.

TABLE II
Failure rate of common components

Component	Failure rate [FIT]	Ref.	Year
Pump laser	53	[40]	2025
Optical switch node	787	[41]	2024
EDFA	10000	[42]	2025
OXC	10000	[43]	2025
Fiber (per 1 km)	100	[44]	2023
Optical line terminal	2500	[45]	2017
ROADM	2100	[46]	2024
Transceiver	4200	[46]	2024
Passive filter	600	[46]	2024
Photodetector	34-102	[47]	2023

A. Optical Components

In optical networks, the main components include transmitters, receivers, OAs, optical cross-connections (OXCs), optical add-drop multiplexers (OADMs), etc. When abstracting actual networks into topologies, OADM or OXC is usually regarded as a node, and optical amplifier (OA) is regarded as part of the link.

Optical components are highly sensitive, and failures often lead to costly, time-consuming repairs. Automated monitoring can detect mechanical faults and accelerate repair if the failure location is known. However, replacements instead of repairs still result in downtime and financial loss. Table II lists failure rates of key optical components, measured in FIT (failures per 10^9 hours). Components like OXCs, OADMs, and OAs are vital to network reliability [48]. This survey focuses on these components and related factors: OADM reliability hinges on structural design, OXC reliability depends on WSS performance and quantity, and OA (e.g., erbium-doped fiber amplifier (EDFA)) reliability depends on load [49]. The following sections detail these influences.

1) OADM: OADM allows specific wavelengths to be added or dropped for local access while passing others through. Traditional OADMs are fixed and pre-configured, lacking the flexibility to conduct wavelength routing dynamically. In contrast, reconfigurable OADMs (ROADMs) support dynamic wavelength add/drop and routing via software featuring colorless, directionless, and contention-less operations. ROADM reliability is intrinsically linked to its architecture. M. Džanko *et al.* systematically compared six ROADM architectures and found that the configuration incorporating N 1:$(N+L-1)$ WSSs, N N:1 couplers, and an OXC achieves the highest reliability, as it optimally balances flexibility and structural robustness [50]. Building on such structural insights, reliability has been further enhanced through intelligent fault localization strategies. R. Wang *et al.* proposed a graph aggregation-based fault screening mechanism that precisely localizes multidimensional node failures while reducing the number of candidate devices by over 98%, significantly improving repair efficiency [51]. To improve cost-effectiveness and scalability under high-load conditions, F. Khosravi *et al.* introduced a low-cost ROADM cluster node design for a three-stage Clos architecture, demonstrating a 3.4% improvement in fiber utilization [52]. Complementing these advancements, S. S. Morrison *et al.* developed a mode division multiplexing-compatible ROADM subsystem featuring mode-selective switching, which achieves -15 dB mode crosstalk suppression and supports 80 Gbps transmission, offering a reliable solution for managing dynamic multimode channels [53]. These efforts collectively reflect the ongoing shift toward more flexible, efficient, and reliable ROADM architectures in modern optical networks.

2) OXC: OXC enables switching between multiple input and output ports and can be implemented in various architectures, such as splitter-based and WSS-based designs or micro-electro-mechanical systems (MEMS)-based structures. In splitter-WSS-based OXCs, each 1×L optical splitter (a passive, highly reliable component [54]) is paired with a 1×L WSS, thus making WSS failures the primary concern regarding reliability. For instance, a 45×45 OXC requires 45 1×45 WSSs, but due to WSS port limitations, each must be split into three 1×20 WSSs, resulting in a total of 270 WSSs for both input and output [55]. Component-level reliability analysis using reliability block diagrams (RBDs) has been applied in [56], accounting for failure/repair rates of splitters, transmitters, receivers, couplers, and multiplexers. Current OXCs support up to 32 wavelengths and operate in the C+L band, typically built using mesh-connected 1×L WSSs. However, future OXCs must support larger capacities (e.g., 64 wavelengths), possibly requiring three-level split-plane switching [57]. [58] introduced a ZnO-doped periodically poled lithium niobate waveguide all-optical wavelength converter with low loss (0.5 dB), high efficiency (10 dB), and strong crosstalk suppression for multi-node transmission. [59] proposed modular designs to reduce hardware complexity and cost. [60] proposes an OXC that reduces cost and loss via optimized configuration. A comprehensive review of OXC structures and reliability is available in [61].

3) OA: OAs compensate for transmission loss but face reliability challenges from aging, nonlinearities, power fluctuations, contamination, and design flaws. They include rare-earth ion-doped amplifiers (e.g., EDFA), nonlinear fiber amplifiers (e.g., Raman), and semiconductor optical amplifiers (SOAs). EDFAs dominate long-haul transmission for their maturity and stability, though high-load stress can degrade reliability [62]. Raman amplifiers offer distributed gain but require high pump power and complex structures. SOAs enable integration and wide bandwidth but suffer from gain instability and thermal effects, impacting reliability.

B. Optical Cables

The reliability of optical cables depends on cable type, environment, and installation method. High-quality materials (low loss, strong, corrosion-resistant) reduce failures, while environmental factors (e.g., temperature, corrosives) impact lifespan. Based on installation, cables are classified as submarine, overhead, or terrestrial, each with specific requirements. The following discusses reliability factors by type.

1) Submarine Optical Cable: Submarine cables carry over 95% of global cross-border data [63], offering long relay distances and stable performance. However, they face mechanical stress from waves, currents, tides, and sea ice, as well as water pressure and corrosion [64]. Protection is provided by armored layers, categorized into light-weight (LW), light-weight protection (LWP), light-weight armored (LWA), single-layer armored (SA), and double-layer armored (DA) [65].

2) Overhead Optical Cable: Overhead optical cables are ideal for stable mountainous and hilly regions. Optical fiber composite overhead ground wire (OPGW) is widely used, offering lightning protection and communication [66]. It comes in three types: stranded stainless steel, center aluminum tube, and aluminum frame. Stranded steel types are compact and support longer fibers. Reliability issues often stem from single-wire breakage, particularly in aluminum alloy strands [67].

3) Terrestrial Optical Cable: Terrestrial cables include direct buried and duct types. Direct buried cables suit stable terrains (plains, hills), while duct cables are used for short distances [68]. Gravity-induced stress and temperature changes can damage cables in steep or undulating regions. Direct buried cables require stronger protection and are armored with steel wire. Their breakage rate is 0.4242/100 km, while duct cables show lower failure at 0.1005/100 km annually [69].

C. Network Topology

In optical network models, components are abstracted as nodes and cables as links, forming a topology $G = (V, E)$, where V and E represent nodes and links, respectively. Common topologies such as linear, ring [70], and mesh [71] are illustrated in Fig. 3. Different topologies suit different applications. Ring topologies enhance local resilience in metropolitan networks, while mesh topologies offer greater fault tolerance in networks [72]. Topology significantly impacts network reliability, especially with a fixed number of nodes and links [73]. Connectivity is a key reliability metric, encompassing node/link connectivity (minimum nodes/links to disconnect the graph) [74]. These metrics are vital under multiple failures, indicating the network's ability to maintain communication [75]. The six topologies in Fig. 3 have connectivity values of 1 to 5, with higher connectivity indicating greater resilience. Low-connected networks are more vulnerable, as failures in key nodes or links can lead to disconnection [76]. Figures 3(d)–(f) show 3-connected, 4-connected, and 5-connected networks, respectively. Accordingly, they remain connected under up to 2, 3, and 4 node failures.

D. Environmental and Human Factors

Most optical components and optical cables are sensitive to the environment, and their reliable operation requires suitable environmental conditions. Environmental changes can impact optical cables or component's service life and failure rate. In some harsh environments, even concurrent multiple failures, cascading failures, network paralysis, and network islands can occur.

1) Atmospheric Pressure: In high-altitude or plateau regions, atmospheric pressure differences significantly impact the performance and longevity of optical components and fibers. Reduced pressure can cause component deformation, sealing failure, and accelerated aging [77]. It also degrades insulation in optical and communication components, increasing electrical failure risks. Additionally, optical cables may experience material expansion or structural deformation, compromising transmission quality and increasing attenuation.

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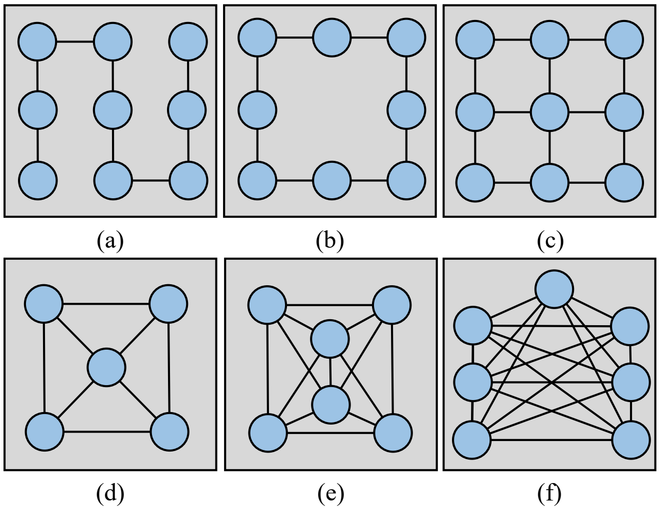

Fig. 3. Different types of topologies, (a) linear, (b) ring, (c) mesh, (d) 3-connected, (e) 4-connected, (f) 5-connected.

2) Dancing: Dancing is a type of aerodynamic instability characterized by large amplitude, low-frequency oscillations of optical cables. This phenomenon typically occurs in overhead optical cables, primarily caused by wind-induced vibrations, and in submarine optical cables, which are affected by ocean currents and tidal forces. For overhead cables, these self-excited vibrations have a low frequency of approximately 0.1-3 Hz and large amplitude, often reaching 20-300 times the optical cable's diameter [78].

3) Temperature: Temperature changes can cause objects to expand and contract at varying degrees, depending on the thermal expansion coefficients of fiber, plastic, and quartz. For example, when the temperature of a fiber unit increases to 59 °C, a strain of 387 $\mu\varepsilon$ is generated, resulting in an additional attenuation of approximately 0.053 dB/km [79]. This attenuation degrades transmission performance, posing challenges for optical networks, especially in areas with significant temperature fluctuations.

4) Ice/snow Cover: Ice/snow cover can affect the performance of optical cables. When ice freezes on an optical cable, it significantly increases the strain on its fiber core, leading to an increase in optical signal attenuation. Once the strain in the fiber exceeds a critical value (typically 0.2% to 0.3%), the fiber will undergo irreversible changes that can severely damage the life of the fiber [80]. In the case of OPGW cables, network capacity can be reduced, and communication interruptions can occur when they are overloaded and disconnected under icing conditions [81].

5) Natural Disasters: Natural disasters cause significant damage to optical networks. For example, Hurricane Katrina in 2005 reduced telecommunications availability from nearly 99.99% to about 85% due to power outages and flooding [82]. The 2008 Wenchuan earthquake in China destroyed 3,897 telecommunication offices, damaged 28,765 km of optical cables, and caused the loss of 142,078 poles [83]. Such failures severely disrupt optical networks, affecting communication system reliability. Natural disasters like hurricanes, fires, lightning, earthquakes, and EMP attacks [84] can damage optical cables, leading to fatigue hardening and reduced service life. Component aging and extreme weather also cause failures, while earthquakes and hurricanes can trigger large-scale disruptions. Correlated failures, such as concurrent (e.g., fiber breaks and component damage) or cascading failures (e.g., power outages causing network blackouts) [85], further complicate recovery. Researchers have explored disaster-tolerant optical networks [86], developing protection and recovery technologies to mitigate these impacts.

6) Human Factors: Human factors in optical network reliability mainly stem from operational errors and construction issues [87]. Operational errors, often occurring during equipment installation, debugging, and configuration, arise when operators fail to follow standard procedures, leading to misconfigurations that significantly impact network reliability. Such errors account for 20% to 70% of optical network failures [88]. Construction issues also pose a major threat, with improper fiber laying or accidental cable damage during excavation and drilling common causes of optical cable cuts [89]. These incidents frequently lead to service disruptions and increased maintenance costs.

E. Failure Types and Failure Rate Modeling

1) Failure Types: Failures in optical networks are categorized as soft failures or hard failures based on their consequences [90]. Soft failures typically do not result in network outages, but the limited number of forward error correction (FEC) failures they induce can degrade transmission performance [91]. Hard failures refer to physical failures in optical components or optical cables, such as component damage or optical cable cuts. Component failures directly degrade the reliability of optical networks. Table III describes commercial optical network failure types. Optical component degradations, controller configuration parameter errors, and dynamic changes can degrade optical network elements [92]. In contrast, inadequate power supply and natural disasters directly result in optical component failures.

2) Failure Rate Modeling: In current reliability studies, a constant failure rate is a common assumption [97]. By assigning fixed failure probabilities to the common failure types listed in Table III, researchers can simplify optical networks' failure modeling and reliability calculation, making it easier to conduct preliminary reliability evaluation. However, this approach overlooks a critical issue: the failure rates of optical components and optical cables are not constant but change over time. For example, the failure rate of a component is typically much higher in the later stages of its service life compared to the early stages of usage.

Probability distribution offers a more accurate description of the characteristics of optical component failures [43]. For example, the aging of components and optical cables may follow a Weibull distribution (also known as the bathtub curve) [98]. Based on the actual probability density functions (PDFs) of optical components and optical cables in operation, the probability density models enhance the accuracy of optical network reliability evaluations.

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TABLE III
OPTICAL NETWORK FAILURE TYPES

Failure Type	Description	Instances
I(Soft)	The physical degradation of components [93]	System fatigue, aging, connector reuse
II(Soft)	Human errors [90]	Parameter misconfiguration
III(Soft)	Unsuitable environment or dynamic changes in components [94]	Fiber stress, power level spikes, increased spectral skew due to EDFA
IV(Hard)	Insufficient power supply leading to optical network failure [43]	Active components will not function properly without power supply
V(Hard)	Human activities [91]	Construction activities
VI(Hard)	Natural disaster-induced downtime [95]	Severe weather and natural disasters
VII(Hard)	Cyber-attacks and terrorist attacks [96]	Telecom infrastructure attacks

F. Reliability Evolution Trends

The reliability of optical components and cables evolves through design, early usage, and late usage (Fig. 4). In the design stage, reliability improves with increased time and cost investment, though later-stage improvements become increasingly costly ($\Delta T_2 > \Delta T_1$). Once a commercially acceptable reliability level is achieved, deployment begins. In early usage, dominated by failure types II, IV, V, VI, and VII, reliability degrades mainly due to random events and harsh environments. Component degradation exists but is not the primary cause of failure. In the later stages, failures related to aging and dynamic effects, such as fiber stress, power spikes, and spectral skew induced by EDFA, become more prominent, leading to higher failure rates [99]. Additionally, human errors, such as configuration mistakes, can occur at any stage.

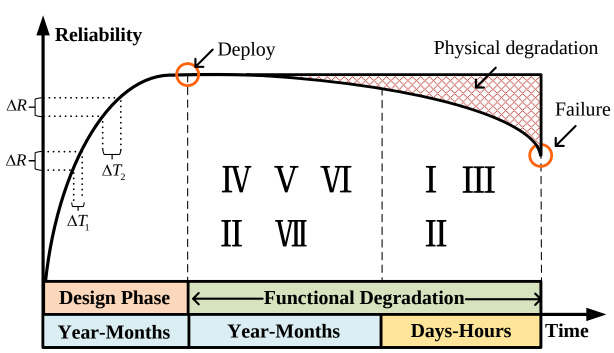

Fig. 4. Time-dependent reliability of optical components and cables.

IV. OPTICAL NETWORK RELIABILITY METRICS

This section outlines the reliability evaluation metrics from three perspectives: optical component/optical cable, service, and topology.

A. Classification of Reliability Metrics

Reliability metrics for optical components and cables include failure rate, repair rate, mean time to failure (MTTF), mean time between failures (MTBF), maintenance-free operating period (MFOP), failure-free operating period (FFOP), and maintenance recovery period (MRP). These metrics quantify reliability based on the frequency and timing of failures. The failure rate is a key indicator that also reflects the overall reliability of the optical network. From a service-oriented perspective, quality of transmission (QoT) and QoS are used to evaluate the transmission quality and operational stability. As discussed in Section III.C, when modeling an optical network as a topological structure, its reliability can be evaluated using graph theory (GT) metrics. The next section provides a detailed introduction to topology-related reliability metrics.

B. Failure Rate and Repair Rate

Failure rate and repair rate are key metrics for evaluating system reliability. In Section III.A, the failure rate is defined as the number of failures per 10^9 operating hours. The repair rate describes the ability to restore a failed device to normal within a unit of time. The repair rate is closely related to the operational capabilities of the maintenance team and the maintainability of the equipment [100]. A higher repair rate indicates shorter downtime and improved system reliability.

C. MTTF, MTTR, MTBF, FFOP, MRP and MFOP

MTTF, MTTR, and MTBF are the most common statistical metrics used to evaluate the reliability of optical networks [101]. Specifically, MTTF reflects the average time a component operates without failures, while MTTR quantifies the average time required to restore a component after a failure. MTBF indicates the average duration between two consecutive failures. A smaller MTTR suggests higher system reliability and faster recovery speed [102], while a larger MTBF implies longer operational intervals between failures and thus higher reliability. In contrast to these statistical indicators, time to failure (TTF), time to repair (TTR), and time between failures (TBF) refer to the observed values in individual failure-repair events. Figure 5(a) illustrates the relationship among TTF, TTR, and TBF.

Another related metric is FFOP, used in evaluating failure-tolerant systems [103]. It measures system degradation during normal operation, ensuring that availability remains within acceptable limits. The Acceptable Level of Degradation (ALD) defines the threshold for tolerable performance decline. MRP involves scheduled inspection and maintenance, while MFOP comprises multiple FFOPs and MRPs, allowing limited errors but preventing overall system failure. Their interrelationship is illustrated in Fig. 5(b).

D. QoT and QoS

QoS and QoT are key metrics for evaluating optical network availability. QoS reflects a network's ability to meet user requirements by managing bandwidth, latency, and reliability at higher layers (e.g., IP or application), while QoT focuses on the physical transmission layer, evaluating the performance of light-paths and light-trees [104]. QoT is typically expressed by bit error rate (BER), optical signal-to-noise ratio (OSNR),

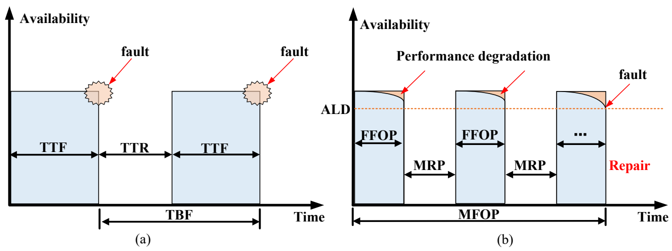

Fig. 5. Relationships between reliability metrics, (a) TTF, TTR, and TBF, (b) FFOP, MRP, and MFOP.

and Q-factor. It is influenced by fiber attenuation, dispersion, nonlinear effects, and inter-channel crosstalk [105]. QoS can be expressed by delay, jitter, and packet loss, and aims to ensure low-latency, high-bandwidth services through traffic prioritization and congestion control [106]. Physical impairments such as ASE noise from EDFAs and crosstalk in OXCs can degrade QoT, increasing delay or jitter and compromising network reliability.

E. Topology-based Reliability Metrics

Topology-based reliability metrics can be divided into structural and central metrics [107]. The structural metrics are primarily studied based on GT and focus on the network's connectivity, i.e., the connectivity when some nodes or links are removed. The structural metrics include connectivity, algebraic connectivity, cut set, etc. On the other hand, the central metrics mainly emphasize the importance of nodes or links and include degree centrality, closeness centrality, etc. Table IV presents the majority of topology-based reliability metrics.

TABLE IV
TOPOLOGY-BASED RELIABILITY METRICS

Metric	Description
Average node degree (Structural)	The mean of all the node degrees in a network [108].
Vertex connectivity (Structural)	The minimum number of vertices that need to be removed to disconnect the graph or make it disconnected [109].
Link connectivity (Structural)	The minimum number of links that need to be removed to disconnect the graph or make it disconnected [110].
Algebraic connectivity (Structural)	The second-smallest eigenvalue (counting multiple eigenvalues separately) of the Laplacian matrix of G [111, 112].
Average shortest path length (Structural)	Average length of shortest paths between all pairs of nodes.
Diameter (Structural)	The longest of all shortest paths between all pairs of nodes.
Assortativity coefficient (Structural)	A measure of the extent to which vertices with the same properties connect to each other [113].
Average two-terminal reliability (ATTR) (Structural)	The ratio of node pairs in each connected component to the network's total number of node pairs. A higher ATTR indicates a more robust network [114].
Cut set (Structural)	A set of nodes/links whose removal disconnects the graph by separating it into at least two disjoint parts [115].
Degree centrality (Centrality)	Degree centrality can be calculated as $D(i)/(N-1)$, where $D(i)$ is the degree of node i and N is the total number of nodes.
Betweenness centrality (Centrality)	The number of times a node appears in all shortest paths [116].
Eigenvector centrality (Centrality)	The entry in the principal eigenvector of the network's adjacency matrix, where the principal eigenvector corresponds to the largest eigenvalue of the matrix [117].
Closeness centrality (Centrality)	The reciprocal of the sum of the lengths of the shortest paths between the node and all other nodes in the graph [118].

V. Reliability Evaluation for Optical Networks

This section begins by classifying optical network reliability evaluation methods into four categories: formula-based, model-driven, data-driven, and other. We will discuss each method's development, fundamental principles, evaluation processes, advantages, limitations, and applicable scenarios.

A. Classification of Reliability Evaluation Methods

Reliability evaluation methods can be categorized by technical approach and theoretical foundation into formula-based, model-driven, data-driven, and other methods, as illustrated in Fig. 6. Formula-based methods rely on mathematical models and statistical theories, using component characteristics such as failure rates or PDFs [119]. Model-driven methods construct and analyze network models to infer risks and identify vulnerable nodes and links [120]. Data-driven methods use historical and indicator data, often employing machine learning (ML) and multi-attribute decision-making (MADM) technologies [121, 122]. This approach is well-suited for small networks with accessible data [123], enabling reliability metric analysis, life estimation, and failure parameter fitting [124]. When data are limited, model-driven methods based on stochastic processes and probability theory are preferable [125]. Other methods include simulation and GT-based approaches. Simulation evaluates reliability by mimicking network behavior under various failure scenarios without disrupting actual operations [126]. GT-based methods evaluate reliability through topological features like connectivity, aiding network pre-design, and identifying critical nodes and failure paths [127].

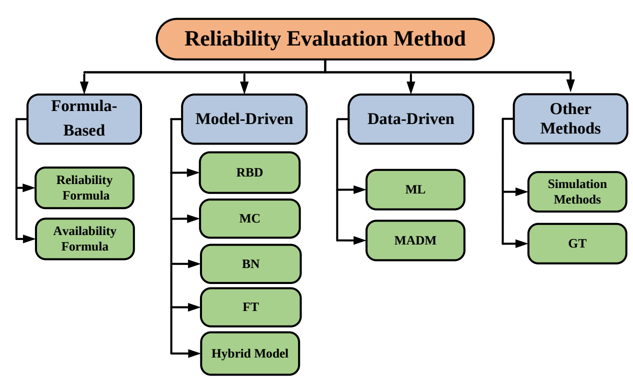

Fig. 6. Summary of reliability evaluation methods.

B. Formula-based Reliability Evaluation

Failures directly reduce the reliability or availability of optical networks. Reliability refers to the probability of stable system operation from $t = 0$ to the end of the observation period without external intervention. Based on this concept, reliability can be calculated using (1), where $R(t)$ denotes the system reliability in $[0, t]$, T denotes the time of failure occurrence, and $P(T > t)$ denotes the probability of failure-free operation in $[0, t]$ [128]. Although many studies assume a constant failure rate to simplify reliability calculations, this is not always accurate. However, if the time between failures follows an exponential distribution, the reliability decreases over time, as shown in (2), where λ represents the system's failure rate. Alternatively, reliability can be calculated using the PDFs of failure events, which describe the likelihood of system failure over continuous time. Based on this approach, reliability can be calculated using (3), where $D(x)$ represents the PDFs of the time between failures.

$$R(t) = P(T > t) \tag{1}$$

$$R(t) = e^{-\lambda t} \tag{2}$$

$$R(t) = 1 - \int_0^t D(x)dx \tag{3}$$

Availability, representing the ratio of available time to total operating time or the probability of operation, can be expressed as (4), where MTTF denotes the mean time to failure, and MTTF+MTTR represents the total observation time [129].

$$A = \frac{MTTF}{MTTF + MTTR} \tag{4}$$

C. Model-Driven Reliability Evaluation

Model-driven reliability evaluation methods are based on pre-established theoretical frameworks and incorporate probability theory, stochastic processes, and statistical analysis. As a result, their effectiveness hinges on the accuracy of prior knowledge and initial assumptions, requiring a comprehensive understanding of the system's structure and parameters. Model-driven reliability evaluation methods encompass RBD [130], Markov chain (MC) [131], Bayesian network (BN) [132] and fault tree (FT) [133]. The choice of model depends on specific requirements, as each model has its unique scope of application. For instance, MC is often utilized in dynamic scenarios with known state spaces, while RBD or FT may be chosen for qualitative analysis based on bistatic theory in static scenarios. Here is an introduction to each model.

1) RBD: The RBD method is widely applied in the design and operational phases of optical network reliability evaluation. It evaluates network reliability based on component failure rates and interconnection structures among optical components [134]. Common connection types include series, parallel, k-out-of-n, series-parallel, and parallel-series, as illustrated in Fig. 7. Among these, the k-out-of-n model represents a redundant system that remains operational if at least k components function properly [135]. The reliability formulas for these five basic RBD models are presented in (5) to (9), where N and M denote the number of components in series and parallel connections, respectively. R_i denotes the reliability of the i_{th} component in a series connection, while R_j represents the reliability of the j_{th} component in a parallel connection. R_{ij} represents the j_{th} parallel component in the i_{th} series component.

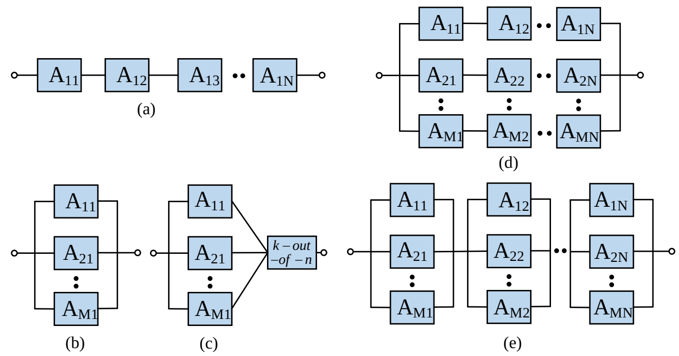

Fig. 7. Five basic structures of RBD, (a) series, (b) parallel, (c) k-out-of-n, (d) series-parallel, (e) parallel-series.

$$R_{series}(t) = \prod_{i=1}^{N} R_i(t) \tag{5}$$

$$R_{parallel}(t) = 1 - \prod_{j=1}^{M} (1 - R_j(t)) \tag{6}$$

$$R_{k-out-of-n}(t) = \sum_{i=k}^{M} C_M^i (R(t))^i (1 - R(t))^{M-i} \tag{7}$$

$$R_{series-parallel}(t) = 1 - \prod_{j=1}^{M} (1 - \prod_{i=1}^{N} R_{ij}(t)) \tag{8}$$

$$R_{parallel-series}(t) = \prod_{i=1}^{N} (1 - \prod_{j=1}^{M} (1 - R_{ij}(t))) \tag{9}$$

RBD is a bottom-up reliability analysis method that helps evaluate the impact of individual component failures on overall network reliability. This method calculates reliability at the component, connection, or network level based on the network's structure. Component-level reliability is determined by analyzing the internal structure of the component [136]. At the connection level, reliability can be calculated using the formula $R = 1-(1-R_{work})(1-R_{backup})$, where R represents the reliability of the optical connection, R_{work} indicates the reliability of working paths, and R_{backup} signifies the reliability of protection paths [137]. Network-level reliability is calculated as the average reliability of all connections within the network [138, 139].

Researchers have also investigated the reliability of ring protection [140] and passive optical networks using the RBD model [141], comparing network reliability under various protection schemes (e.g., path protection, span protection, and p-cycle) [142–144], as well as in scenarios involving dual failures [145]. Nonetheless, the RBD model has certain limitations in reliability analysis. For instance, it cannot handle time-varying or complex failure types and is restricted to fixed failure types and rates [146].

2) Markov Model: The Markov model is a stochastic model with Markovian properties, widely used for reliability evaluation, failure diagnosis, and lifespan analysis of complex networks [147, 148]. Compared with RBDs, Markov models more effectively capture dynamic state transitions, including time-dependent failures and repairs, and better represent optical network components' dependencies. Markov models include MCs and Markov decision processes. MCs analyze system reliability using state transition diagrams and matrices, where components transition between states with defined probabilities [149]. This enables transient and steady-state analyses of complex interactions and dependencies [150]. Input parameters typically include failure and repair rates, while outputs indicate the system's state probabilities or steady-state availability [151].

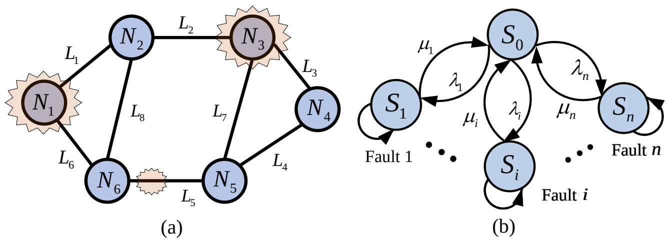

Fig. 8. A single-failure Markov model, (a) a 6-node optical network with eight links, (b) Markov state transfer diagram.

An illustration of a 6-node optical network with eight links and its corresponding Markov state transfer diagram is shown in Fig. 8. The state set of the optical networks is $S=\{S_0, S_1, \cdots, S_{14}\}$, where the network has a total of $(1+6+8)$ states when only a single failure is considered. The optical network typically operates in state S_0, while states S_1 to S_{14} represent the failure states. Furthermore, there will be $(1 + A_{14}^2)$ states when only considering dual failures. Among them, S_0 represents the normal state of the system, while $S_1 - S_{182}$ represents various combinations of dual-component failures. λ_i and μ_i denote the failure rate and repair rate of component i, respectively. The analysis of the Markov process can be simplified into four steps, taking a single failure as an example. The state transfer matrix can be derived from (10), which allows the calculation of the probability that the optical network is in a specific state at any moment based on its initial state (transient analysis). Specifically, the probability of being in the S_0 state represents the current availability of the optical network. By applying (10)-(11), the probability of the system reaching a steady state as time approaches infinity can be determined, which is referred to as steady-state analysis.

$$\frac{dP_i(t)}{dt} = -\sum_{j=0, j\neq i}^{N} a_{ij} P_i(t) + \sum_{k=0, k\neq i}^{N} a_{ki} P_i(t) \tag{10}$$

where a_{ij} represents the rate of transition from state i to state j, while a_{ki} denotes the rate of transition into state i from state k. If state i represents the normal state and state j represents the failure state, then a_{ij} represents the transition rate from the normal state to the failure state, i.e., the failure rate, which can be denoted by λ_i. N represents the total number of failure states. $P_i(t)$ represents the probability of being in state i at the moment t.

$$\lim_{t\to\infty} \frac{dP_i(t)}{dt} = 0, \quad i \in \{0, 1, \cdots, N\} \tag{11}$$

Furthermore, the single-failure scenario can be extended to a dual-failure scenario [152]. A Markov state transition diagram for the dual-failure scenario is depicted in Fig. 9. Here, S_0 represents the normal state of the system, S_N denotes a single failure, and $S_{N,N-1}$ signifies a dual failure. Subsequently, the Markov state transition equation can be expressed as (12). Here, P_0 represents the probability of normal state of the optical network, P_i signifies the probability of component i being in a failure state, and P_{ij} denotes the probability of component j failing given the failure of component i. Notably, P_{ij} and P_{ji} are not equivalent. P_{ji} denotes the probability of component i failing given the failure of component j.

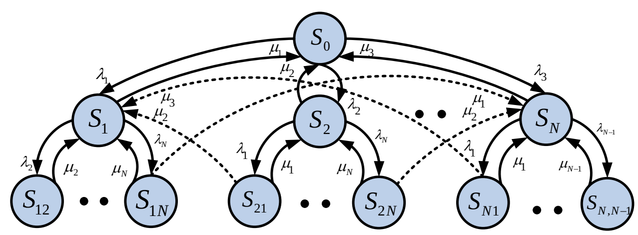

Fig. 9. Markov state transfer diagram for dual-failure scenarios [152].

$$P_i\mu_i + \sum_{j=1, j\neq i}^{N} P_i\lambda_j - P_0\lambda_i - \sum_{j=1, j\neq i}^{N} (P_{ij} + P_{ji})\mu_j = 0 \tag{12}$$

Here, $P_i\mu_i$ represents the transfer probability from state P_i to state P_0, $P_i\lambda_j$ represents the transfer probability from state P_i to state P_{ij}, $P_0\lambda_i$ represents the transfer probability from state P_0 to state P_i, $(P_{ij} + P_{ji})\mu_j$ denotes transfer probability from state P_{ij} or state P_{ji} to state P_i. By utilizing (12) to (14), the instantaneous reliability of the network at any given moment and the steady-state reliability as time approaches infinity can be calculated.

$$P_{ij} = \frac{\lambda_j P_i}{\mu_i + \mu_j} \tag{13}$$

$$P_0 + \sum_{i=1}^{N} P_i + \sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} P_{ij} = 1 \tag{14}$$

Two primary Markov-based approaches are widely used for optical network reliability evaluation [153]. The first focuses on modelling the state transitions of network components, capturing the dynamic behaviour of the system [152, 154]. Some studies refine this approach by defining state spaces based on failure sequences or incorporating backup resource utilization through sequential Markov models to estimate availability [155, 156]. However, these models often face scalability limitations, with a limited demonstration of their applicability to large-scale optical networks. Another commonly adopted strategy models key reliability-influencing factors as binary variables [157, 158]. For example, network reliability can be represented through three binary dimensions: connectivity, average BER, and the proportion of unavailable links. This abstraction allows for the construction of compact Markov models with eight distinct states, enabling a structured yet simplified evaluation framework.

3) BN: BN is a probabilistic graphical model for capturing the conditional independence between variables. It is widely used in the quantitative system reliability evaluation [159]. BN consists of directed acyclic graphs (DAGs) and conditional probability tables (CPTs) [160]. The directed links represent the conditional dependencies between nodes, while the CPT models the dependencies between random variables. Static Bayesian networks (SBNs) and dynamic Bayesian networks (DBNs) are common types of BN. In an SBN involving N random nodes $A_1, A_2, \cdots, A_N$ and a directed acyclic graph, the joint distribution probability is denoted as (15). Here, $\mathrm{Pa}(A_i)$ represents the parent set of node A_i. $P(A_i|\mathrm{Pa}(A_i))$ represents the probability that node A_i takes a specific value given the states of its parent nodes $\mathrm{Pa}(A_i)$.

$$P(A_1, A_2, \cdots, A_N) = \prod_{i=1}^{N} P(A_i|\mathrm{Pa}(A_i)) \tag{15}$$

Bidirectional reasoning leverages the conditional independence assumption and the chain rule of BN. Forward reasoning is used for reliability evaluation, while reverse reasoning supports failure diagnosis [161]. Reasoning methods are classified as approximate or exact: approximate reasoning (e.g., sampling or belief propagation) offers computational efficiency at the cost of accuracy, whereas exact reasoning (e.g., variable elimination or conditional reflection) provides precise results but requires substantial computational resources [162]. SBN applies when the subject's reliability is time-independent. However, SBNs are inefficient for large-scale networks and cannot capture dynamic behavior among variables [163]. DBN extends SBNs over time, including a prior network B and a transition network $B_{\rightarrow}$. As shown in Fig. 10, intra-slice arrows represent dependencies within a time slice, while inter-slice arrows reflect temporal dependencies between variables across slices.

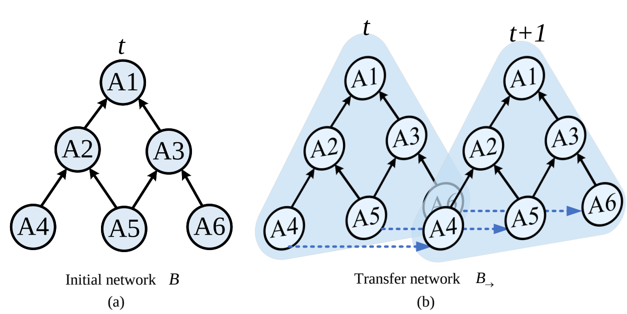

Fig. 10. DBN, (a) initial network, (b) transfer network.

By expanding the time slices, the joint distribution probability can be obtained [164]. It can be expressed as (16), where $\mathrm{Pa}(A_t^i)$ represents the parent set of node A_i at time t. $P(A_t^i|\mathrm{Pa}(A_t^i))$ represents the probability that node A_i takes a specific value given the states of its parent nodes $\mathrm{Pa}(A_t^i)$ at time t. N denotes the number of states in the network at time t (i.e., the number of nodes in the state diagram).

$$P(A_{1:n}^{1:T}) = \prod_{t=1}^{T}\prod_{i=1}^{N} P(A_i^t|\mathrm{Pa}(A_i^t)) \tag{16}$$

BN has increasingly been explored for reliability evaluation in optical networks, offering a probabilistic framework that integrates expert knowledge and empirical data. By combining these information sources, early work such as [165] demonstrated improved prediction accuracy. Subsequent studies expanded BN's applicability and leveraged prior knowledge to evaluate reliability under multi-link failures [166], while [167] introduced a subsystem-based modeling approach to identify critical components. BN has also been applied to optimize multi-link protection technologies, enabling differentiated reliability provisioning [168]. More recently, time-dependent BN models have been proposed to incorporate system degradation and enhance dynamic reliability estimation [169]. Using BN in optical network reliability remains a promising yet evolving research direction.

4) FT: FT analysis is a graphical technique used for evaluating the reliability of complex systems [170]. It is characterized by strong logical reasoning capabilities, supporting qualitative and quantitative assessments. The analysis starts from the top event (i.e., system failure) and traces back through logical gates to uncover the underlying causes. By identifying the minimum cut sets, FT analysis can enumerate all potential failure states within the network [171]. A typical FT consists of a top event, basic events (representing root causes), intermediate events (between basic and top events), and logic gates connecting these events. Fig. 11(a) illustrates the main logic gates and event symbols used in FT.

FT is constructed by analyzing various causes of network failures, with the failure event defined as the top event. It identifies all contributing factors, including intermediate and basic events, and connects them via logic gates. Subsequently, the minimum cut set, representing the path leading to network failure, is identified, and the failure probability of the top event is calculated using probabilistic methods. In Fig. 11(b), T represents a failure event, M_1–M_6 depicts intermediate events, and x_1–x_5 represents basic events. Based on the logical relations, the Boolean operations describing the occurrence of the T event yield the simplest formula: $T = x_1x_5 + x_3x_4 + x_2x_4x_5 + x_1x_2x_3$. Consequently, the minimum cut sets of failure event T are determined as $S_1 = \{x_1, x_5\}$, $S_2 = \{x_3, x_4\}$, $S_3 = \{x_2, x_4, x_5\}$, and $S_4 = \{x_1, x_2, x_3\}$. In FT analysis, reliability is typically defined as the probability that none of the cut sets corresponding to failure events occurs, expressed as (17), where $P(S_i)$ denotes the probability of cut set S_i occurring.

$$R = \prod_{i=1}^{n} (1 - P(S_i)) \tag{17}$$

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5) Hybrid Model-Driven Reliability Evaluation: Given the complexity of network topologies, inter-component dependencies, and diverse failure modes, no single modeling approach can fully capture the reliability characteristics of modern optical networks [172]. For instance, FT analysis excels at identifying causes of failure but lacks dynamic and probabilistic reasoning capabilities. Consequently, FT is often integrated with models like MC or BN, where FT defines the system's failure structure and MC or BN performs probabilistic inference, enabling a more comprehensive evaluation. To address challenges such as multi-type failures, dynamic traffic patterns, and structural complexity, hybrid modeling has emerged as a promising strategy [173]. These models offer greater flexibility and scalability by combining complementary techniques, particularly for large-scale, distributed, and time-varying networks [174]. Recent studies underscore the potential of hybrid approaches in optical network reliability evaluation. Y. Guo *et al.* integrated dynamic DBNs with XGBoost, using DBNs to capture component dependencies and XGBoost to infer real-time states from high-dimensional monitoring data [175]. S. Rebello *et al.* combined DBNs with hidden Markov models (HMMs), where HMMs transformed continuous observations into hidden state probabilities for DBN-based inference [176]. Z. Chen *et al.* proposed a hybrid MC-BN framework to model dynamic transitions and environmental uncertainties jointly [177]. These examples highlight how hybrid models leverage the strengths of different techniques to deliver more accurate and adaptive reliability evaluations.

6) Evaluation Processes: Figure 12 outlines the construction process of four fundamental reliability evaluation models, comprising three main phases: system modeling, reliability evaluation, and validity verification. The network structure, component failure rates, and state transitions are defined in the system modeling phase. The evaluation phase involves quantitative or qualitative analysis based on model structure and metrics such as failure-free time, minimum cut sets, or failure probabilities. In the validity phase, sensitivity analysis is first conducted to evaluate how input variations affect outcomes, followed by scenario effectiveness evaluation and model adjustment for decision-making.

7) Advantages and Limitations: Model-driven approaches are limited by reliance on rational assumptions and precise parameter settings. RBD provides a clear visual of system-component relationships, making it suitable for quickly analyzing simple systems. MC captures state transitions, enabling dynamic continuous and discrete system analysis. BN models complex dependencies and supports fuzzy and precise reasoning under uncertainty, making it effective for complex systems. FT identifies root causes of failures via minimal cut-set analysis. However, RBD cannot model dependencies or dynamic behavior and is limited to simple systems. MC assumes constant failure rates and suffers from state space explosion in large systems. BN requires extensive prior knowledge and high computational cost for accurate inference. FT struggles with systems involving redundancy or priority logic and depends on substantial prior data.

8) Characteristics and Applications: Table V summarizes the characteristics and applicable scopes of four common reliability evaluation models. The BN and MC support multi-state systems, offering high flexibility and dynamic adaptability, making them suitable for quantitative and qualitative analysis in the post-design stages. However, they rely heavily on prior knowledge and computational complexity. In contrast, the RBD and FT models are relatively simpler, fitting quantitative and qualitative analysis during the design and operational phases. Although less flexible, their inference capabilities and complexity are more moderate. The RBD model requires minimal prior knowledge, whereas the FT model necessitates some understanding of the optical network structure and failure mechanisms.

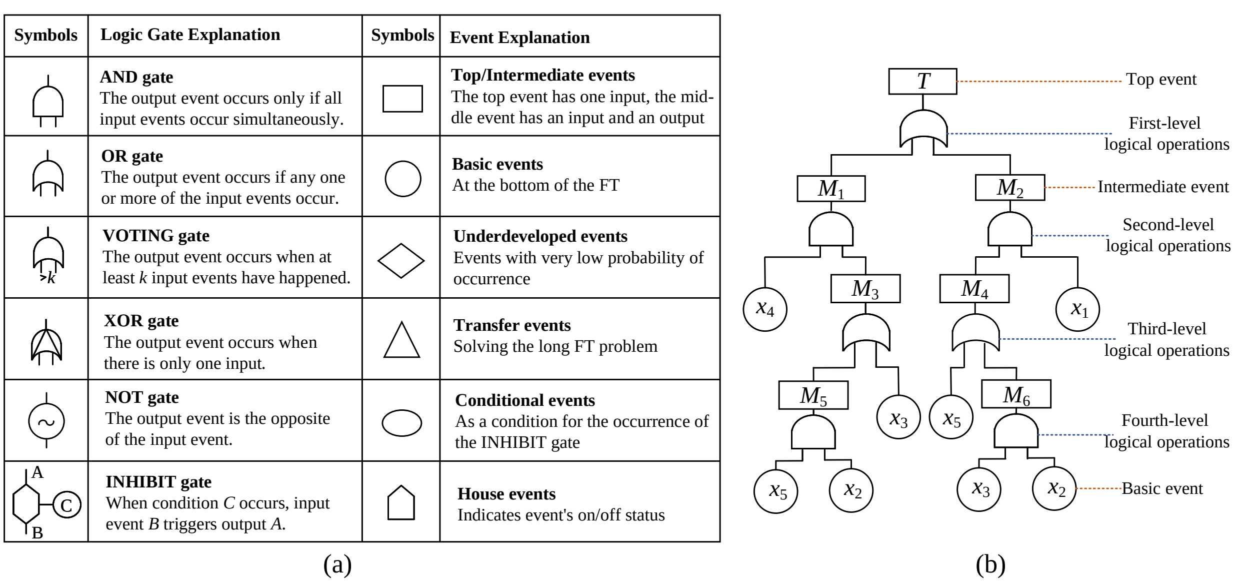

Fig. 11. FT analysis method, (a) basic logic gate symbols and event symbols in FT, (b) structure of the FT analysis.

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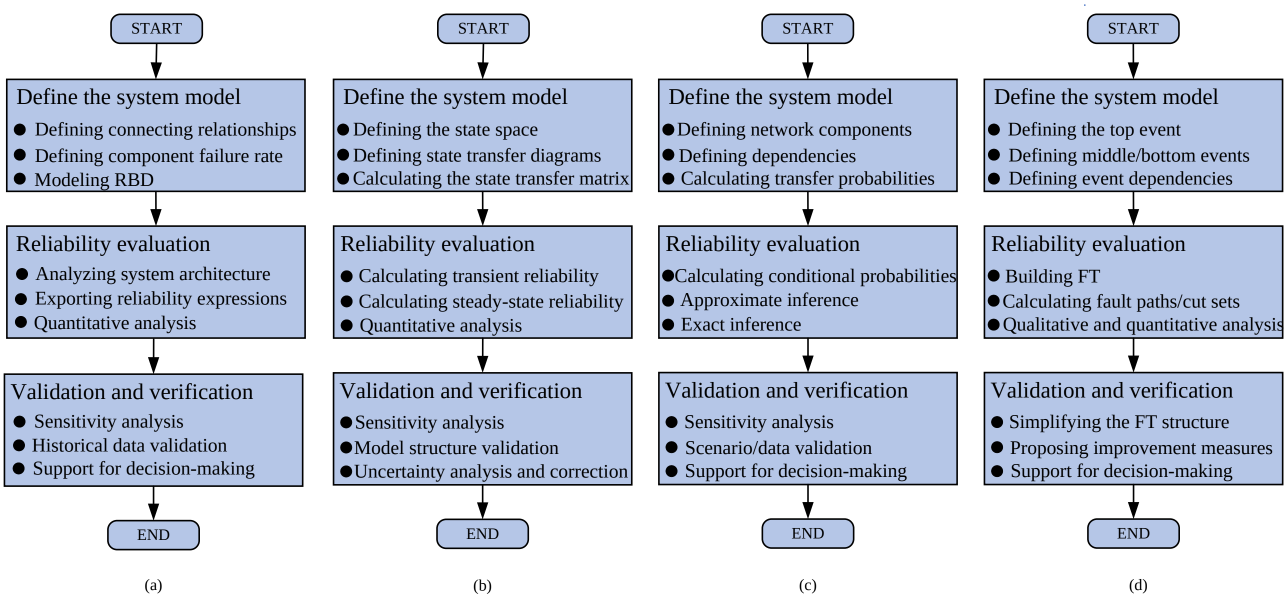

Fig. 12. The evaluation processes of building four reliability evaluation models, (a) RBD-based reliability evaluation method, (b) MC-based reliability evaluation method, (c) BN-based reliability evaluation method, (d) FT-based reliability evaluation method.

TABLE V
CHARACTERISTICS AND SCOPE OF APPLICATION OF RELIABILITY EVALUATION MODELS

Model	State space	Flexibility and dynamic	Quantitative	Qualitative	Phase	Prior knowledge	Computational complexity	Reasoning power	Dependency modeling
BN	Support polymorphism	High	Yes	Yes	Post-design	Required	High	High	Available
MC	Support polymorphism	High	Yes	Yes	Post-design	Required	High	Medium	Available
RBD	Bistable	Low	Yes	Yes	Operation and design	Unnecessary	Low	Low	Unavailable
FT	Bistable	Low	Yes	Yes	Operation and design	Required	Medium	Medium	Unavailable

D. Data-Driven Reliability Evaluation

Traditional approaches in optical network reliability evaluation have primarily relied on model-driven methods due to the difficulty of obtaining data. In contrast, data-driven methods leverage monitoring data collected from various network devices. By processing these monitoring data, data-driven evaluation methods can evaluate the operational states of key network components, enabling early detection of potential failures or performance degradation. This section focuses on the application of ML in optical network reliability evaluation, specifically exploring its role in evaluating service availability, predicting the reliability of nodes and links, and supporting MADM.

1) ML-Based Evaluation Methods: QoT prediction is a widely used data-driven method for evaluating optical network service reliability. Accurate predictions help network managers evaluate performance before path setup, enabling timely adjustments and enhancing operational reliability. Algorithms such as deep neural networks (DNNs) predict QoT based on service parameters like hop count, modulation format, and transmission rate [178–180]. Convolutional neural networks (CNN) have also been used for intelligent OSNR estimation from eye diagrams [181]. Enhanced ML strategies have emerged to tackle challenges like limited field data and varying transmission conditions. In [182], a feature selection method with transfer learning improved QoT prediction under data scarcity. A deep learning classifier predicted whether BER in single-mode fiber (SMF) or few-mode fiber (FMF) links would exceed thresholds [183]. An invariant CNN model in [184] incorporated a fixed-length encoder and robust predictor to handle variable link configurations, improving QoT estimation accuracy. Beyond light-path prediction, data-driven approaches have also been applied to multicast services. Studies in [185] used DNNs to evaluate light-tree availability. For an in-depth review of ML-based QoT prediction, refer to [186].

For component reliability evaluation, Y. Chen *et al.* employed a temporal convolutional network to evaluate the reliability of optical network components by analyzing historical data and multidimensional sensor inputs [187]. Similarly, [188] proposed a DNN-based digital twin model that transforms component data into health scores, enabling performance degradation prediction and supporting timely maintenance or replacement to ensure network continuity.

2) MADM: MADM quantifies the metrics influencing the reliability of an optical network [189]. A set of metrics and their corresponding weights is typically utilized during multi-attribute evaluation, reflecting the relative importance of various metrics in the overall evaluation. Various methods are available for researchers to analyze and evaluate the reliability of optical networks. For instance, techniques for order preference by similarity to an ideal solution (TOPSIS) [190], analytic hierarchy process (AHP), and entropy weight method (EWM) allow for the systematic comparison of different network designs and configurations [191, 192].

E. Other Methods

1) Simulation Methods: In reliability evaluation, simulation-based methods, such as Monte Carlo simulation (MCS), subset simulation, and weighted sampling, offer effective ways to handle system uncertainty and complexity [193–195]. Among them, MCS is the most widely adopted due to its robustness and flexibility [196]. It enables detailed modeling of optical network failures (e.g., fiber cuts, OA failures) by generating statistical data on metrics like availability, delay, and transmission quality through repeated sampling. Compared to model-driven methods, MCS offers two major advantages: it accommodates realistic failure behaviours via probability distribution functions (e.g., Weibull) [197], and it avoids state explosion by using simple statistical computations, making it scalable to large systems [198]. Recent studies demonstrate MCS's applicability in various contexts. For example, X.G. Chen *et al.* used event tables to simulate network state changes [199]. R. Vaisman *et al.* explored shared risk link group (SRLG)-based failure scenarios [200]. Agent-based modelling has emerged as an effective method for capturing run-time interactions in optical networks [201]. The agent-based method is another type of simulation method that evaluates network reliability through component states. By employing intelligent agents, the agent-based method can represent the dynamic behaviors of network components, including aging, failures, repairs, functional changes, and the propagation of faults across the network [202]. Through multi-agent cooperation, the agent-based method can represent state transitions and help maintain system robustness in complex, time-varying environments. For instance, a five-layer, agent-based architecture was proposed, in which optical and power-grid elements are modelled as coupled agent layers [43]. Layer-to-layer interactions enable the framework to quantify how aging, random failures, and interconnect structure affect optical network reliability.

TABLE VI
RELIABILITY EVALUATION METHODS BASED ON GT

Topic	Ref.
Evaluation frameworks for topological robustness	[203–205]
Under different attack scenarios (random/targeted)	[206]
Resilience evaluation (post-failure performance)	[207, 208]
ML-driven robustness prediction	[209, 210]
Minimum cut-set based reliability computation	[211, 212]
Binary decision diagrams (BDDs) approach	[213, 214]
Fast binary addition tree	[215]
Geographical constraint-aware topology optimization	[216–219]
Generalized topological robustness design frameworks	[220, 221]

2) GT-Based Evaluation Methods: Reliability evaluation in optical networks using GT focuses on evaluating the topology's robustness [203]. Key metrics for this evaluation include node degree, cut set, and degree centrality, with network connectivity being the most crucial. Connectivity analysis checks if a network can maintain communication despite node or link failures, ensuring at least one path exists between any pair of nodes [204]. GT-based reliability analysis helps in the early design phase, allowing network designers to evaluate and optimize the robustness of topologies [205]. Table VI presents studies on the reliability and robustness of optical network topologies.

F. Evaluation Processes, Advantages, and Limitations of Data-Driven Methods and Other Methods

1) Evaluation Processes: As shown in Fig. 13, the evaluation processes of four methods generally follow three stages: model construction, reliability evaluation, and optimization. ML-based methods involve data preprocessing, model training with algorithms like decision trees or neural networks, and reliability prediction for network optimization. MADM-based approaches focus on selecting and weighting multiple reliability metrics, constructing a standardized decision matrix, and scoring options to guide optimization. Simulation-based methods model system structure and activities using probability distributions and then evaluate reliability through time-series analysis. GT-based approaches abstract the network as a graph, analyze failure propagation, and apply graph algorithms to evaluate and enhance structural robustness.

2) Advantages and Limitations: ML-based methods leverage reliability data to predict service and component reliability in optical networks (e.g., light-paths and light-trees). These methods are flexible and effective but depend heavily on data quality and quantity and often require significant computational resources for training and optimization. MADM-based methods support multi-criteria decision-making by integrating multiple attributes and expert knowledge. Their strength lies in comprehensively considering various reliability factors and assigning weights based on importance, enabling more precise evaluations. However, the subjectivity in weighting and the complexity of constructing standardized decision matrices can introduce bias and reduce accuracy. Simulation-based methods use probabilistic modeling to capture network behavior and failure dynamics under diverse scenarios, offering realistic and detailed evaluations. Despite their high flexibility, they are often time-consuming and computationally demanding, especially in large-scale networks. GT-based methods offer a structured approach for analyzing network topology and critical nodes, effectively evaluating robustness and failure propagation. However, their focus on structural reliability limits their capacity to evaluate overall network performance comprehensively.

3) Characteristics and Applications: Table VII summarizes four reliability evaluation models and their applicability. ML-based methods in optical networks mainly predict service transmission quality and component lifespan. They are flexible and dynamic but rely heavily on prior knowledge, such as

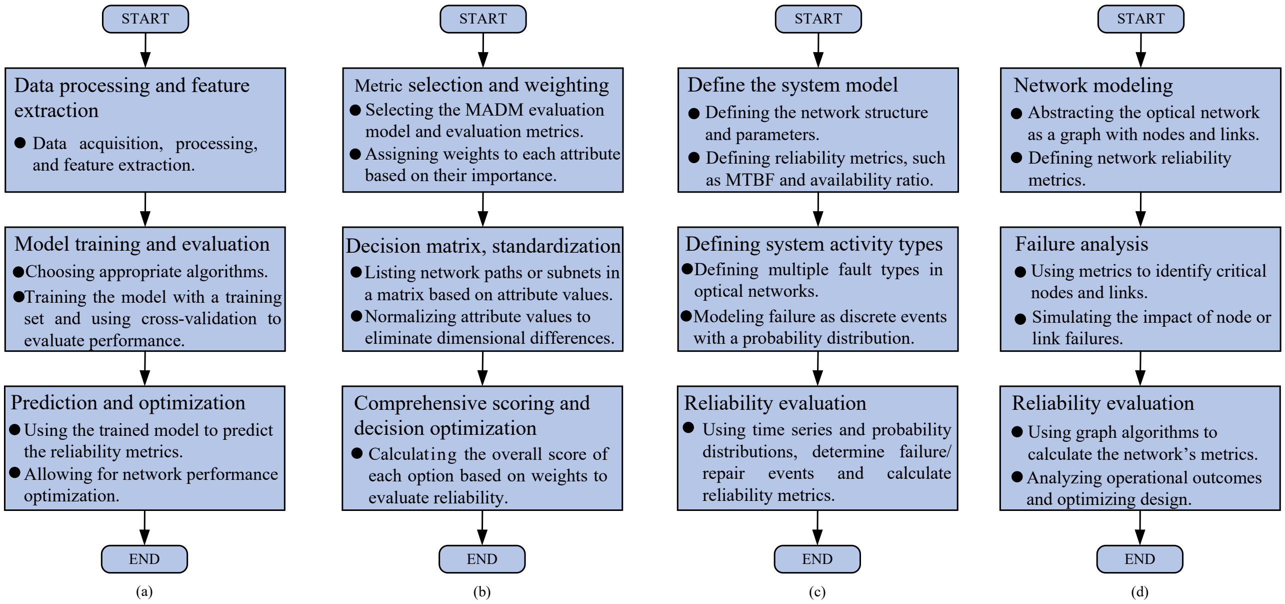

Fig. 13. The evaluation processes for the four reliability evaluation methods, (a) ML-based reliability evaluation method, (b) MADM-based reliability evaluation method, (c) simulation-based reliability evaluation method, (d) the GT-based reliability evaluation method.

TABLE VII
CHARACTERISTICS AND APPLICATION OF THE DATA-DRIVEN RELIABILITY EVALUATION METHOD AND OTHER METHODS.

Method	State space	Flexibility and dynamic	Quantitative or qualitative	Phase	Prior knowledge	Computational complexity	Reasoning power	Dependency modeling
ML-based	Support polymorphism	High	Both	Deployment	Required	High	Low	Available
MADM-based	Support polymorphism	Medium	Both	Design and deployment	Required	Medium	Medium	Limited
Simulation-based	Support polymorphism	High	Both	Design and pre/post Deployment	Unnecessary	High	Medium	Available
GT-based	Bistable	Low	Both	Design	Unnecessary	Low	Low	Available

failure data and performance metrics. The MADM approach suits the design and deployment stages, balancing flexibility with the ability to analyze both quantitative and qualitative factors, though its inference capacity is limited. Simulation-based methods support reliability evaluation across the design and deployment lifecycle, simulating network structures, maintenance, and protection technologies. These methods offer high flexibility and computational capability with less dependence on prior data. GT-based methods, focused on optimizing low-reliability topologies, are best applied in the design phase and feature lower computational complexity.

VI. Methods to Improve the Reliability of Optical Networks

Survivability technologies that accelerate post-failure recovery substantially improve the reliability of optical networks. This section investigates survivability technologies applied to optical networks. To begin with, general survivability technologies applicable across different types of optical networks are introduced. Subsequently, specific survivability technologies tailored to particular network types are elaborated. Disaster recovery technologies and differentiated reliability (DiR) schemes are also summarized. Finally, the reliability evaluation of certain protection technologies is introduced.

A. Optical Network Survivability Technologies

Enhancing optical network reliability relies on protection and recovery technologies, which differ based on resource reservation. Protection schemes precompute and reserve spare resources (routes and wavelengths) before a failure occurs, ensuring rapid and complete failure recovery at the cost of inefficient capacity utilization [222]. In contrast, recovery schemes dynamically discover spare resources upon failure, offering higher resource utilization and flexibility but with longer recovery times and no guarantee of complete failure recovery [223]. Industry standards, such as those specified in SLAs, often mandate that recovery times for a single failure should not exceed 50 milliseconds [224].

1) Optical Network Recovery: Recovery in optical networks refers to the dynamic computation and configuration of recovery paths after a failure [225, 226]. This process is carried out using the remaining available network resources, and may not succeed if the network is resource-constrained [227]. Recovery technologies can be classified according to operation scope and implementation framework.

According to the operation scope, recovery can be categorized into link-based recovery, path-based recovery, and sub-path-based recovery [228]. Path-based recovery is an end-to-end rerouting technology that establishes a new path between the source and destination nodes. Its advantage lies in enabling global resource optimization, thereby improving overall resource utilization efficiency [229]. However, this comes at the cost of longer signaling propagation and path computation delays. Link-based recovery performs localized reconnection between the adjacent nodes of the failed link, replacing only the affected link. This method achieves very fast recovery speed, but since it relies solely on local information, it often results in relatively low resource utilization efficiency [230]. Sub-path-based recovery serves as a compromise between the two technologies. With a recovery granularity between link and path recovery, it reroutes the affected sub-path (rather than just a single link or the whole path), thereby striking a desirable balance between recovery speed and resource utilization efficiency.

According to the implementation framework, recovery can follow either a distributed or a centralized model [231]. Early optical networks relied on manual provisioning via network management systems, which was slow and error-prone. A control plane was later introduced to automate connection setup, most notably with generalized multiprotocol label switching (GMPLS) [232]. In GMPLS, recovery involves several collaborating function modules [233]. The link management protocol (LMP) is responsible for detecting faulty links and maintaining link connectivity information. Once a failure is identified, open shortest path first with traffic engineering (OSPF-TE) extensions, which disseminate the updated topology and resource state across the network. Based on this information, the head-end router uses the constrained shortest path first (CSPF) algorithm to compute working and backup paths that satisfy specific constraints (e.g., SRLG) [234]. Subsequently, the resource reservation protocol with traffic engineering (RSVP-TE) extensions reserves link resources and conducts cross-connections along the computed route, thereby setting up a recovery path. In a centralized model, a software-defined networking (SDN) controller computes a recovery path after a link or node failure. Within the controller, the routing module performs path computation, and the signaling process is realized through a southbound protocol such as the OpenFlow, thereby establishing an end-to-end recovery path [235, 236].

2) Optical Network Protection: Ring and mesh protection schemes are the two main types of network protection. Bellcore introduced the SONET standard, which standardized self-healing ring architectures such as unidirectional path-switched rings (UPSR), unidirectional line-switched rings (ULSR), two-fiber bidirectional line-switched rings (BLSR/2), and four-fiber bidirectional line-switched rings (BLSR/4) [237]. Each node has a single alternate path in a ring topology, enabling fast failure recovery and enhanced resilience, but this requires 100% spare capacity for protection [238]. As network sizes and traffic demands grew, mesh networks gradually replaced ring-based architectures [239]. Mesh networks offer better capacity utilization and routing flexibility. It needs only 40%-60% extra capacity for shared path protection [240]. Compared with recovery technologies, protection technologies provide a more direct and readily measurable improvement in network reliability [26]. Furthermore, they encompass a broader range of approaches, differing in protection scope, sharing principle, and structure. The subsequent section offers a detailed discussion of these protection schemes.

B. Protection Scheme

Protection schemes vary widely and can be classified based on different criteria. For instance, in terms of protection scope, they include path protection, link protection, and sub-path protection. When categorized by resource allocation, they can be divided into shared protection and dedicated protection, depending on whether backup resources are shared. Schemes such as p-cycle and pre-configured polyhedron protection can also be identified based on the structural arrangement of protection resources. The following sections will provide a detailed discussion of each scheme.

1) Path Protection: Path protection involves rerouting traffic to a backup path when a link fails on the primary path, substituting the primary path with the backup one [241]. The links on the primary and backup paths must not intersect to ensure reliability. For instance, in Fig. 14(a), the primary path is $A\to B\to E\to H$, while the backup path is $A\to D\to G\to H$. If a failure occurs on link $B\to E$, node E detects the failure and sends an alarm signal to the source node A via a dedicated signaling channel. Upon receiving the alarm, node A redirects traffic along the backup path $A\to D\to G\to H$.

2) Link Protection: The protected link is provided with a dedicated backup path in link protection. This approach enables faster switching, as only a switching signal needs to be transmitted between the two endpoints of the failed link. However, it comes at the cost of reduced rerouting flexibility [242]. For example, in Fig. 14(b), the primary path is $A\to B\to E\to H$. If a failure occurs on link $B\to E$, traffic is redirected to the protection route $B\to C\to F\to E$.

3) Sub-Path Protection: Sub-path protection, a compromise between path and link protection, divides the primary path into segments, each individually protected [243]. While this requires dedicated backup resources for each segment, it allows for rapid recovery through localized operations. As shown in Fig. 14(c), the primary path $A\to B\to E\to H$ has the protected sub-path $B\to E\to H$. Suppose a failure occurs in $E\to H$, node H alerts node B, and with pre-assigned backup paths (e.g., $B\to C\to D\to G\to H$). In that case, recovery is initiated quickly, bypassing the need for end-to-end path recomputation and rerouting traffic within milliseconds.

4) Dedicated Protection: Protection mechanisms are divided into dedicated and shared protection, depending on whether backup resources are shared. Dedicated protection reserves exclusive resources, including 1+1 and 1:1 schemes. In 1+1 protection, traffic is sent simultaneously over both working and backup paths, with the receiver selecting the better signal [244]. In 1:1 protection, the backup path carries additional traffic under normal conditions but switches to carry primary traffic upon failure. While 1+1 offers faster recovery, it requires double the capacity, increasing cost. In contrast, 1:1 is more resource-efficient but incurs longer recovery due to path switching [245].

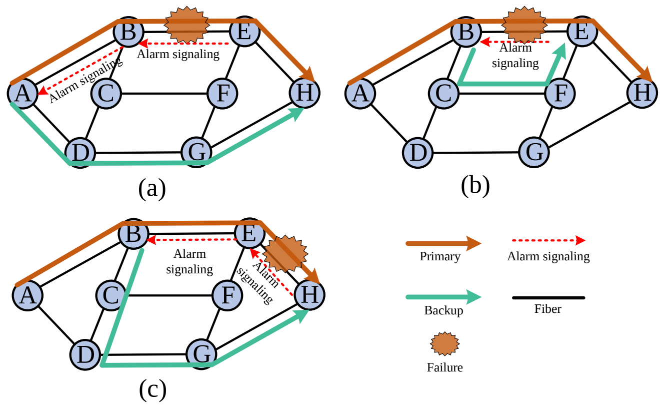

Fig. 14. Principles of different protection schemes, (a) path protection, (b) link protection, (c) sub-path protection.

5) Shared Protection: Shared protection allows for sharing backup resources only if their working paths are independent or SRLG-independent. This ensures that no resource contention occurs during a failure. SRLG refers to links sharing a common physical resource, such as optical cables, which may result in multiple fiber breaks during fiber cuts [246]. In shared protection, switches cannot be configured on a path before a failure, leading to longer recovery times and increased complexity compared to dedicated protection. This is because shared protection requires additional connection-sharing information [247].

M: N protection falls under the category of shared protection. Each working link is associated with multiple backup links in the M: N protection scheme. Specifically, there are N working links and M backup links, with $N \geq M$. When a working link fails, the system quickly recovers by rerouting the affected data traffic from the failed link to a backup link [239]. M: N protection offers superior failure tolerance as it allows multiple backup links for each working link, reducing the impact of multiple failures. In contrast, a 1: N protection scheme associates each working link with only one backup link. While this scheme is simpler and less costly than M: N protection, it may provide lower failure tolerance [26].

Shared backup path protection (SBPP), proposed by the Internet Engineering Task Force (IETF), is a failure-independent, path-based protection scheme [248]. It establishes separate working and backup paths, allowing backup channels from different working paths to share backup resources [249]. Unlike M:N protection, SBPP does not require paths to share the same sender and receiver. Its advantages include efficient capacity sharing and failure-independent protection, simplifying control by avoiding failure localization [250]. As a path-based scheme, SBPP reroutes traffic entirely via the backup path regardless of the failure point [244, 251].

6) P-cycle: P-cycle is a pre-configured protection method proposed by D. Stamatelakis *et al.* in 2000 [252]. A link-based protection approach combines a ring network's rapid switching capability with a mesh-based network's capacity efficiency [253]. The method establishes a ring structure by pre-connecting the network's backup capacity. The p-cycle can protect both on-cycle links and straddling links, providing double the protection capacity for straddling links [254]. As illustrated in Fig. 15(a), consider a cycle with spare link capacity, represented as $N_1 \to N_2 \to N_4 \to N_5 \to N_7 \to N_6 \to N_3 \to N_1$. When a failure occurs on the on-cycle link $N_2 \to N_4$, the spare capacity available on the p-cycle can provide protection. In the case of a straddling link failure, such as the link $N_4 \to N_6$, although this link is not part of the p-cycle, both N_4 and N_6 are endpoints of the p-cycle. As a result, two backup paths are available, ensuring dual protection, as shown in Fig. 15(b).

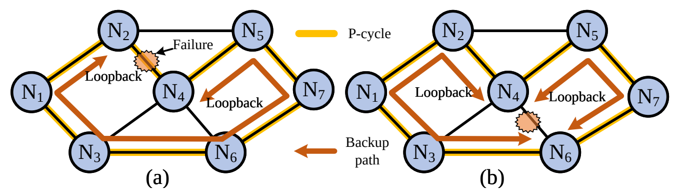

Fig. 15. Pre-configured protection structure, (a) a failure on an on-cycle link, (b) a failure on a straddling link.

The p-cycle method faces several limitations. It protects only cyclic and cross-links, often requiring larger cycles to cover more nodes and links, which may extend backup paths, slow recovery, and degrade optical signals [255, 256]. Restricting p-cycle size necessitates more cycles, increasing management complexity [257]. Designing p-cycles involves regenerator placement, path length, and modulation formats. Evaluating these aspects helps select optimal p-cycle sets for protection [258]. Existing studies explore p-cycle availability and optimization, progressing from dual-failure to multi-failure scenarios to enhance network protection [256–258].

The p-cycle concept has been extended to path/subpath protection, leading to the failure-independent path-protection (FIPP) p-cycle [259]. Unlike traditional link-specific protection, FIPP protects entire working paths and enables real-time failure detection without relying on direct connections to the end nodes [260]. It provides independent end-to-end path switching while maintaining the p-cycle integrity [261]. The FIPP p-cycle assumes bidirectional optical paths and reroutes traffic if a link or node fails, using a spare capacity p-cycle. FIPP is more capacity-efficient than traditional p-cycles but has a slower recovery speed [262]. FIPP p-cycles have also been applied to EONs, where H. M. N. S. Oliveira *et al.* developed algorithms for single- and dual-failure protection, and further introduced traffic grooming and spectrum sharing between disjoint backup paths to enhance spectrum efficiency [263]. In addition, they extended the FIPP p-cycle to SDM-EONs and proposed multigraph-based routing, spectrum, and core allocation (RSCA) algorithms, considering inter-core crosstalk [264].

7) Pre-Configured Polyhedron Protection: Traditional protection and recovery methods face survivability and resource efficiency limitations under multiple link failures [265]. The cube protection strategy extends the p-cycle concept into three-dimensional space. For example, using k-regular and k-(edge) connected structures offers a robust protection scheme based on Menger's theorem. It ensures k edge-disjoint paths between any two nodes, maintaining network operation despite up to $k-1$ simultaneous link failures [266]. This approach effec-

This article has been accepted for publication in IEEE Communications Surveys & Tutorials. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/COMST.2025.3621165

tively addresses concurrent failures while optimizing backup resource usage, enabling rapid recovery with reduced spare capacity requirements [267]. In such k-edge-connected structures, $m/(d-m)$ gives the recovery capacity's lower bound, where d is the average node degree and m denotes failed components.

8) Multipath Protection: Multipath-based protection schemes transmit traffic from source to destination via multiple paths. Data is split into low-speed streams, each on a separate path. If one path fails or is overloaded, others continue to support degraded but uninterrupted service [268]. As shown in Fig. 16, a 6 Gbps connection from node A to H uses sub-wavelength granularity [269]. Fig. 16(a) uses two 3 Gbps paths, allowing degraded service upon a single failure. Fig. 16(b) shows a 2:1 protection setup ensuring continuity with reduced performance. Multipath protection supports fine-grained resource allocation, including spatial granularity, reduces blocking via small frequency gaps, and improves reliability [270].

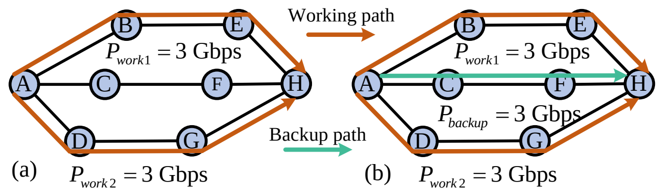

Fig. 16. Multipath protection, (a) the traffic is split into 3 Gbps for dual-path transmission, (b) the traffic is split into 3 Gbps for dual-path transmission with 2:1 protection.

Research on multipath protection focuses on six key areas. First, adaptive schemes dynamically allocate paths based on traffic to improve resource utilization [271]. Second, optimizing residual capacity involves leveraging unused bandwidth and pre-planning protection paths to boost service availability [272, 273]. Third, integration with network functions virtualization (NFV) enables multipath-based virtual network function (VNF) placement, improving flexibility and manageability [274, 275]. Fourth, load balancing and path optimization can enhance performance [276, 277]. Fifth, configuration sensing helps reduce service degradation risks [278, 279]. Finally, joint management of differential delay constraints (DDC) and QoS ensures consistent performance across diverse paths [280].

C. Survivability Technologies for SDH Networks

In SDH networks, self-healing rings and automatic protection switching (APS) are crucial in ensuring network reliability [281]. Self-healing rings provide an efficient and practical protection mechanism, enabling rapid recovery from node and link failures. These rings can restore failed services within 50 milliseconds, making them highly effective in maintaining network reliability [30]. Several types of self-healing ring architectures, including UPSR, ULSR, and BLSR, can operate with two or four fiber links (BLSR/2 and BLSR/4, respectively). On the other hand, APS is designed to handle typical link failures and protects different switching mechanisms based on resource allocation strategies.

1) UPSR: UPSR comprises two optical fibers. As depicted in Fig. 17(a), fiber T is the primary transmission path, with fiber P as a backup path. The service is transmitted from node A to node D for reception, with the receiving end selecting the superior quality signal. In case of interruption in the connection between E and D (as illustrated in Fig. 17(b)), the channel switch redirects the transmission from fiber T to fiber P to receive signals transmitted by the latter. Once the failure is rectified, the switch will revert to its initial position. This mechanism boasts a rapid switching speed (<50 ms). However, its drawback lies in its limited capacity and incapability to transmit additional services.

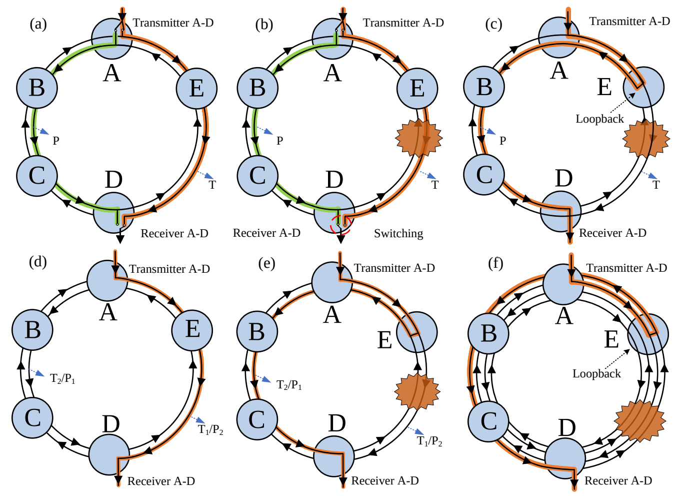

Fig. 17. Self-healing ring protection technology, (a) UPSR working normally, (b) failure in $E \rightarrow D$ of UPSR, (c) failure in $E \rightarrow D$ of ULSR, (d) BLSR/2 working normally, (e) failure in $E \rightarrow D$ of BLSR/2, (f) failure in $E \rightarrow D$ of BLSR/4.

2) ULSR: ULSR comprises two optical fibers. During normal, signals are solely transmitted through fiber T, while backup fiber P remains inactive. Each node is outfitted with a switch preceding the branch signal-tapping function. In the event of a disconnection between E and D, depicted in Fig. 17(c), the protection inversion switches at nodes E and D execute the APS protocol loopback function. At node E, the signal transmitted from node A is rerouted from fiber T to fiber P, traversing through nodes A, B, and C counterclockwise before reaching D. Node D receives the signal from fiber P.

3) BLSR/2: In a BLSR/2 architecture, two counter-rotating fiber rings each allocates 50% capacity for working traffic and 50% for protection, enabling bidirectional redundancy. Under normal conditions (Fig. 17(d)), a signal from node A travels through nodes A, E, and D via the working channel T_1 on the T_1/P_2 fiber, with P_2 as its protection. Similarly, T_2 and P_1 serve as the working and protection channels on the other fiber. When a failure occurs between nodes E and D (Fig. 17(e)), node E loops the signal from T_1 to P_1, which then travels via nodes A, B, and C to D. At node D, the signal loops back from P_1 to T_1 for delivery. This channel reuse improves capacity, but the APS protocol and crossover delay remain limitations.

4) BLSR/4: BLSR/4 enhances BLSR/2 by using four fibers in two counter-rotating pairs, with separate fibers for working and protection traffic, ensuring full physical isolation and greater capacity. In normal operation, signals from node A pass through E to reach D, while the backup fiber stays idle. If a failure occurs between E and D (Fig. 17(f)), nodes E and D perform loopback to reroute traffic onto the protection fiber.

D. Survivability Technologies for WDM Networks

In WDM networks, survivability technologies include optical multiplexing segment (OMS) layer protection and optical channel (OCh) layer protection [282]. The main difference is their scope of protection. OMS protection focuses on recovering all services from a failed link by rerouting traffic through backup paths. Common schemes include 1+1 OMS protection, 1:1 OMS protection, and OMS-dedicated or shared protection rings. OCh protection operates at the level of individual optical channels, assigning backup channels to restore affected ones. Representative OCh protection schemes include 1+1 OCh dedicated protection, OCh-Shared Protection Ring, and OCh Mesh Protection. These protection mechanisms enhance resilience and ensure continuous service in WDM networks.

E. Survivability Technologies for EONs

EONs were proposed to enable more flexible spectrum utilization. In an EON, the spectrum on fiber links is divided into frequency slots (FSs). Typical ITU-T flexible-grid parameters use a 12.5 GHz nominal slot width with 6.25 GHz central-frequency granularity [283]. As defined by ITU-T for fixed-grid WDM, channel spacings are fixed (e.g., 100 GHz or 50 GHz) [284]. In contrast, EONs dynamically allocate multiple contiguous FSs to match service demands, improving spectral efficiency.

1) Bandwidth Squeezing Restoration (BSR): BSR is an effective survivability technique for EONs [285]. Unlike traditional protection schemes that aim to recover affected traffic fully, BSR focuses on protecting or restoring only a portion of the primary traffic during a failure [286]. In conventional optical networks, failures disrupt optical paths and degrade bandwidth. BSR minimizes this impact by dynamically compressing bandwidth, enabling partial service recovery under failure conditions. Although BSR cannot fully recover the bandwidth of the failed connection, it ensures the continuity of critical services [287].

2) Split Spectrum Approach (SSA): In EONs, optical channels can be partitioned into multiple sub-bands using technologies such as SSA [288] and virtual concatenation [289], enabling transmission across diverse routes. This approach significantly enhances network reliability and flexibility by distributing sub-bands over multiple paths. During failure recovery, these sub-bands can be dynamically restored via different routes, improving the network's fault tolerance [290]. By leveraging virtual connection mechanisms, spectrum resources can be efficiently utilized across multiple paths, increasing the overall resilience of the network. Despite failures, degraded service can still be maintained at the receiving end, preventing complete service interruptions [291].

F. Survivability Technologies for SDM Networks

With the continuous growth of internet traffic, SMF's capacity limitations are becoming increasingly evident. SDM technology has been introduced in optical networks, leveraging the spatial dimension to enhance spectral resource utilization and improve the capacity of optical fiber channels [292]. MMF and MCF are the primary infrastructures supporting SDM. By utilizing different transmission modes or multiple cores within a single fiber, SDM enables multiple spatial channels to coexist, significantly increasing system capacity [293]. These channels operate independently at distinct wavelengths with minimal interference, optimizing spectral efficiency. Currently, no dedicated survivability mechanisms have been specifically designed for SDM networks. However, existing research has explored several primary survivability technologies for SDM, including dedicated protection [294], shared protection [295], compression-based protection [296], and multipath protection [297]. These technologies aim to enhance fault tolerance and ensure service continuity in SDM-based optical networks.

G. Survivability Technologies for IP Over WDM Networks

IP over WDM is a network architecture that transports IP data over WDM optical networks to improve efficiency and performance [298]. Failure recovery and traffic aggregation are managed at the optical layer, while the IP layer provides an additional recovery mechanism for multi-layer failures. The IP layer's key advantage is its ability to differentiate traffic flows, enabling targeted failure recovery for specific data streams [299]. However, single-layer recovery technologies have limitations. While the WDM layer handles some failures, it cannot address all disruptions or provide differentiated protection. The IP layer offers flexibility but is limited by high control complexity and slower recovery.

1) Failure Propagation: In multi-layer networks, a key challenge is failure propagation across layers. Failures in bottom-layer nodes or links can render multiple top-layer links unavailable, leading to inter-layer disruptions [300]. As shown in Fig. 18(a), optical layer nodes (A–F) correspond to IP layer nodes (A_1–F_1). A failure in the optical link ($C \rightarrow E$) may impact multiple IP links such as ($C_1 \rightarrow E_1$) and ($A_1 \rightarrow E_1$), which belong to the same SRLG. Thus, IP-layer recovery must handle multiple simultaneous logical link failures. In multilayer networks, recovery techniques can be implemented following three different strategies. [301].

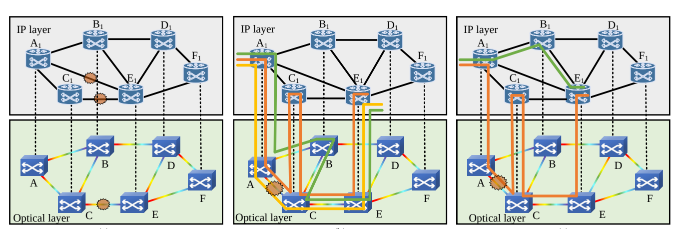

Fig. 18. Coordination mechanisms for multiple-layer protection, (a) failure propagation, (b) bottom-layer recovery, (c) top-layer recovery.

2) Bottom-Layer Recovery: Bottom-layer recovery involves failure recovery close to the source of the failure. Figure 18(b) illustrates two services: the first traverses $A_1 \rightarrow C_1 \rightarrow E_1$ (via logical links $A_1 \rightarrow C_1$ and $C_1 \rightarrow E_1$), while the second is a direct link from A_1 to E_1. An optical fiber cut in the optical layer ($A \rightarrow C$ failure) affects both services at the IP layer (logical link $A_1 \rightarrow C_1$ and $A_1 \rightarrow E_1$). In the optical layer, recovery can be achieved through a link protection mechanism. For example, the $A \rightarrow B \rightarrow C \rightarrow E$ route can recover service. However, suppose optical node C experiences a failure. In that case, the first service cannot be resolved in the optical layer because IP layer node C_1 is isolated due to the failure at optical layer node C [301]. Conversely, the second service can be addressed using a bottom-layer recovery scheme.

3) Top-Layer Recovery: In survivability technologies provided by the IP layer, failure recovery is implemented close to the service. For example, as illustrated in Fig. 18(c), consider a service from A_1 to E_1. When a failure occurs in the optical layer ($A \rightarrow C$ fiber cut), recovery at the IP layer can be achieved by routing the traffic via nodes $A_1 \rightarrow B_1 \rightarrow E_1$. In the optical layer, there are multiple path options for implementing the logical link $A_1 \rightarrow B_1 \rightarrow E_1$, such as paths $A \rightarrow B \rightarrow C \rightarrow E$ and $A \rightarrow B \rightarrow D \rightarrow E$.

H. Survivability Technologies for Data Center Optical Networks

With the advent of cloud computing, content-centric networking has emerged as a promising approach to improve data center reliability against large-scale failures. By replicating content across multiple nodes, data centers enhance service availability and adopt flexible traffic models such as anycast and multicast [302, 303]. These models allow users to access services from any replica, offering improved fault tolerance and resource efficiency. Unlike traditional node-to-node connectivity, content connectivity enables service continuity even when direct paths fail, as users can retrieve data from any accessible replica [304]. In optical networks, conventional protection mechanisms may fail if all source-destination paths are disrupted. However, content replication ensures uninterrupted access as long as at least one path to any data center remains available [305]. To support reliability under this model, research has focused on virtual network mapping schemes [306], optimal data center placement strategies [307, 308], content partitioning and placement technologies [309, 310], and robust backup planning [311, 312]. These technologies collectively enable resilient, content-aware optical infrastructures.

I. Disaster Preparedness Measures

Disaster recovery is vital to optical network reliability, involving proactive and reactive measures to maintain essential services and data security during catastrophic events [313]. Its importance is heightened in unpredictable situations like natural disasters, cyberattacks, or large-scale technical failures, which often cause extensive infrastructure damage. For instance, earthquakes can sever fiber links, floods may submerge base stations, and hurricanes might destroy transmission towers, severely disrupting network operations [314]. While traditional protection schemes effectively handle single or dual failures, they often fall short during large-scale disasters [315]. Events like cascading failures can disrupt power supplies, affecting optical components, while vertically correlated failures may propagate from the optical to the IP layer, causing data loss or service outages. These complex scenarios require more adaptive and comprehensive recovery technologies [316].

Although large-scale disasters are relatively rare, provisioning excessive backup resources is often impractical and costly. As a result, dynamic recovery has become a key strategy in disaster response [317]. Unlike static backups, it enables real-time resource reallocation and path adjustments based on actual damage, supporting rapid recovery of critical services. This includes real-time monitoring, quick identification of affected areas, dynamic backup allocation, and, when necessary, deployment of temporary communication facilities [318]. In addition to dynamic rerouting, various technologies further enhance resilience. Content connectivity leverages replicated data in distributed centers to ensure access despite failures [319, 320]. Risk-aware strategies use disaster risk maps to guide network design and routing, reducing exposure in high-risk zones [321, 322]. Service degradation strategies dynamically adjust resource use based on latency, bandwidth, and availability requirements, ensuring continuity under constraints [323].

J. DiR

DiR is a protection paradigm that aligns optical network reliability mechanisms with varying service demands and priorities [324]. By assigning different protection or degradation levels to connections based on their reliability and importance requirements, DiR supports hierarchical reliability provisioning. A widely adopted classification scheme relies on service recovery time [325], typically dividing services into high, medium, low, and best-effort tiers. This approach balances cost and reliability by allocating more protection resources to critical services while offering limited protection to non-critical ones [326]. For example, financial and medical applications require near-zero downtime, whereas delay-tolerant services like email can tolerate best-effort reliability. Service classification in DiR is often based on interruption thresholds, as summarized in Table VIII. Two commonly referenced thresholds are 50 milliseconds and 2 seconds [325]. Services demanding sub-50 ms recovery benefit from mechanisms such as permanent 1+1 protection, ensuring uninterrupted performance except in extreme transient conditions. In contrast, services with tolerable delays of up to 2 seconds can withstand brief disruptions, such as minor frame loss in video streams, without significantly affecting user experience.

TABLE VIII
SERVICE RELIABILITY CLASSIFICATION

Reliability level	High	Medium	Low	Best effort
Recovery time	<50 ms	<500 ms	<2 s	<60 s
Protection methods	Dedicated protection	Shared protection	Recovery	Recovery

Differentiated protection and configuration strategies, aiming to tailor protection schemes according to service importance and resilience requirements [324, 325]. The second research direction addresses configuration degradation, which dynamically adjusts protection levels and resource allocations in response to failures or resource limitations, thereby maintaining service continuity under adverse conditions [327, 328].

K. Reliability Evaluation of Protection Technologies

Various protection and recovery technologies have been proposed to enhance optical network reliability, yet their effectiveness requires systematic evaluation. Existing research primarily focuses on evaluating protection technologies and verifying their ability to meet user reliability demands. For instance, studies have shown that 2:1 protection can achieve higher reliability levels (e.g., four nines) [329]. Agent-based models and heuristic methods have also been employed to evaluate failure-repair behaviors and connection availability under different protection schemes [43, 330]. However, results vary significantly due to differences in metrics and simulation settings [331]. Nevertheless, a key objective of reliability evaluation methods is to identify vulnerable links within optical networks and optimize protection technologies by balancing reliability and cost [332].

VII. Reliability: Balancing Energy Consumption and Costs

Section VI shows that survivability technologies can greatly lower the chance of service interruptions and enhance reliability. However, they add extra fibers, ports, amplifiers, and other hardware, increasing daily operation and maintenance costs and energy consumption. Furthermore, they reserve additional spectrum, reducing sellable capacity and thus potential revenue. This section summarizes two key trade-offs researchers and operators face: reliability versus cost, and reliability versus energy consumption. It reviews where cost and energy arise in optical networks and explores how to cut both while meeting reliability requirements.

A. Reliability vs. Cost

For telecom operators, the primary challenge lies not in achieving revenue growth per se but in sustaining service-level guarantees under increasingly constrained financial conditions. Recent industry outlooks report that global fixed-mobile service revenue increased by 4.3% in 2023, reaching US$1.14 trillion. However, growth is expected to decelerate to a compound annual growth rate (CAGR) of only 2.5% through 2028 [333]. At the same time, reliability requirements defined in telecom reliability standards remain stringent [334]. Achieving “five-nines” availability and sub-50 ms recovery typically requires consuming a large amount of redundant resources, substantially enhancing network reliability but inevitably increasing overall costs [335]. Therefore, it is very necessary to balance reliability and cost, ensuring compliance with reliability requirements while minimizing expenses.

Network costs are generally divided into capital expenditure (CAPEX) and operational expenditure (OPEX) [336]. CAPEX includes long-term investments in infrastructure, such as fiber deployment, equipment procurement (including routers, switches, ROADM/OXC, and optical amplifiers), installation and commissioning, and site construction costs. OPEX covers daily operation expenses, including staff employment, equipment maintenance and repair, energy consumption (which usually accounts for a large proportion of OPEX), site rental, software licenses, troubleshooting, and emergency response costs. The composition and relationship between reliability and total cost are shown in Fig. 19. A reasonable allocation of both is essential for ensuring sustainable network development and profitability [337].

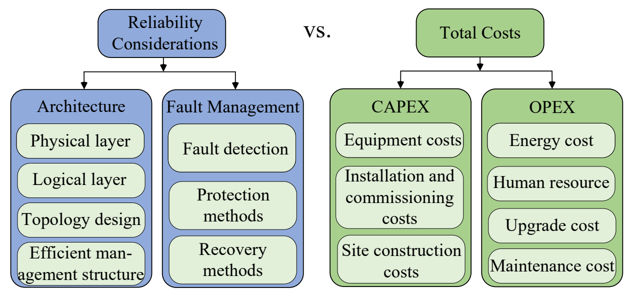

Fig. 19. The composition and reliability considerations of optical network expenditure.

In the full lifecycle of optical networks, the relationship between cost and reliability shows clear dynamic changes. During the planning and construction stages, the focus is on the reasonable allocation of CAPEX. On the one hand, it is necessary to reduce early failure rates through topology design, redundant links, modular ROADM/OXC design, hierarchical fiber layout, and geographic site layout that fully considers terrain and harsh environmental factors [338]. On the other hand, although redundant configurations can improve reliability, excessive redundancy will significantly increase upfront investment, future maintenance costs, and equipment depreciation. During the operational stage, cost shifts to OPEX, including daily maintenance, energy consumption, software licensing, and emergency responses [339]. High reliability requirements may increase OPEX due to reserved backup resources and active monitoring.

Studies have explored a wide range of strategies to balance reliability and cost in optical networks, demonstrating that high reliability can be achieved without imposing excessive expenses. These strategies span four dimensions of network design and operation: topology optimization, component reduction, and management framework improvement, which apply to the planning and construction stages, and preventive maintenance, which applies to the operational stage. First, topology optimization is effective in enhancing both throughput and reliability. For example, traffic-aware bi-connected topologies can improve network reliability without increasing total fiber length [340]. Second, reducing the number of network elements has been a central focus. Genetic algorithms have been applied to minimize line interfaces and switches in transport networks, yielding optimal designs [341], while heuristic scheduling approaches further cut the number of required line cards [342]. Third, management frameworks are key in balancing cost and reliability. Cost-effective management frameworks, particularly those incorporating hybrid management planes, optimize resource utilization and enable real-time performance-driven recovery technologies. Such frameworks can achieve failure recovery within 50 ms while lowering CAPEX by an average of 8%, and up to 40% in long-haul networks [336, 338]. Finally, preventive maintenance has emerged as a practical alternative to costly emergency responses. Preventive maintenance transforms unexpected failures into manageable events by employing delayed repair schemes that schedule early interventions. For example, AI-driven monitoring tools enable early fault prediction and resource optimization, thereby reducing the reliance on costly emergency repairs and minimizing manual intervention expenses [343, 344]. Collectively, these approaches underscore that carefully coordinated design and operational strategies can simultaneously ensure robust reliability and economic efficiency in modern optical networks.

B. Reliability vs. Energy Consumption

In modern communication networks, energy consumption has become an issue that cannot be ignored. High reliability usually depends on deploying redundant links to maintain sufficient backup capacity. However, this inevitably increases the amount of equipment and energy consumption. For example, in carrier backbone networks, 1+1 or 1:1 protection mechanisms are widely used to prevent single failure, requiring double the amount of fibers and equipment, significantly raising energy consumption. Conversely, improving network energy efficiency requires increasing optical devices' utilization rate and load level, which may negatively affect network reliability [345]. Ericsson projects that the overall electricity consumption of the information and communication technology sector will rise by about 10% between 2023 and 2030, reaching around 1,100 TWh. Within this total, networks are expected to grow by 6% [346]. This increasing energy demand for optical networks translates into a fundamental trade-off: reducing power consumption while maintaining high reliability. As illustrated in Fig. 20, existing research can be broadly categorized into static and dynamic methods.

Static methods are primarily applied during the network planning and construction stages, focusing on selecting low-power and highly reliable components, designing energy-efficient topologies, and developing protection schemes that balance energy efficiency with reliability [347]. First, choosing low-power and highly reliable components directly lowers the overall energy consumption of the network [348]. Second, energy-efficient topology design has been extensively studied. Research shows that optimizing the physical topology by jointly considering link deployment and nodal degree can achieve significant energy savings in optical backbone networks while maintaining reliability [349]. Mixed-integer linear programming (MILP) evaluations further indicate that such designs can reduce power consumption by around 10% under normal traffic loads and up to 90% under extreme scenarios [350]. Third, protection schemes are critical because they determine the amount of redundant resources that must remain active. More redundancy resources usually improve reliability but increase energy use, while more efficient schemes reduce redundant resources and thus save energy. For example, replacing the energy-intensive but highly reliable 1+1 protection with 1:1 protection can cut optical-layer energy consumption by up to 49% [351]. Similarly, allocating DiR levels to traffic demands based on their SLAs can reduce backup resources and improve energy efficiency. Compared with conventional 1+1 protection, DiR achieves up to 25% energy savings [352]. In multilayer networks such as IP-over-WDM, placing protection at the optical layer and adopting a top-down design methodology can reduce power consumption by as much as 65.7% compared with uncoordinated dual-layer schemes [353]. These studies highlight that static methods always involve a trade-off: reducing energy consumption while ensuring network reliability remains acceptable.

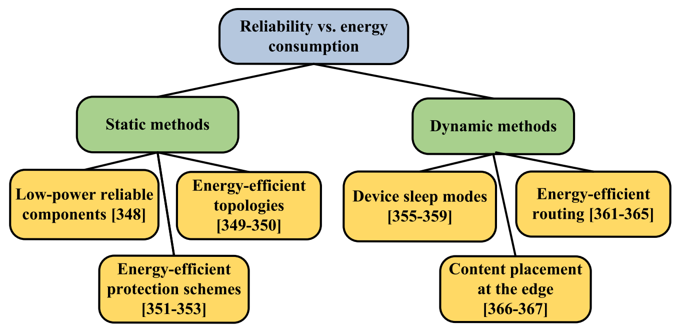

Fig. 20. Reliability-aware energy-efficient methods.

In contrast, dynamic methods focus on management and maintenance strategies during the operational stage [354]. Typical approaches include scheduling device sleep modes, computing energy-efficient routes, and placing content at the network edge. These methods improve energy efficiency but always involve a trade-off between short-term savings and long-term reliability. First, optical network components can enter sleep modes during low-load or idle periods without affecting reliability, provided that wake-up latency stays within QoS and protection switching limits [355]. MILP-based and heuristic-based scheduling of line card sleep has significantly cut power consumption [356, 357]. However, frequent on–off cycles speed up device aging and threaten service reliability. To reduce this risk, some schemes cap the number of daily transitions and integrate device lifetime models into scheduling, ensuring that sleep is activated only when savings energy outweigh reliability costs [358]. Modular routers provide another option by allowing only backup line cards to sleep while keeping primary ones active. This preserves reliability while saving energy [359]. Still, repeated power changes create thermal stress on chips, power modules, and solder joints, raising reliability risks as sleep strategies become deeper [360]. Second, green routing integrates power consumption directly into routing and wavelength assignment decisions. To realize energy-efficient routes, common formulations use integer linear programming (ILP) or heuristic algorithms that explicitly enhance efficiency by minimizing the number of active links and regenerators [361–363]. Beyond pure energy-aware formulations, Natalino *et al.* proposed an energy- and fatigue-aware routing strategy that models component switching-induced aging to balance energy savings with connection reliability [364]. However, in large-scale network topologies, exact ILP models and even sophisticated heuristics are computationally heavy, making it difficult to select energy-efficient routes quickly and thereby increasing routing computation time, undermining reliability. Therefore, reinforcement learning–based approaches to energy-efficient routing have been proposed, which compute energy-efficient routes while simultaneously considering computation time and reliability. For example, Q-learning combined with heuristic graphs can discover energy-efficient routes while reducing routing computation time [365]. Finally, placing content at the network edge reduces transport-network energy consumption by serving requests over shorter paths and with fewer active devices [366], while multi-tier replication preserves availability during wide-area failures [367]. Overall, dynamic methods highlight a clear trade-off: achieving energy savings at the acceptable cost of reduced device lifetime, service stability, or QoS continuity.

VIII. Experience and Future Work

This section first outlines the overall key lessons and insights. Then, it focuses on the application experiences, improvement measures, and future development directions of three existing typical evaluation models. Subsequently, it discusses development directions for optical network reliability evaluation, including multi-dimensional evaluations based on reliability, availability, maintainability, and safety (RAMS), AI-driven methods, digital twin-assisted technologies, integrated closed-loop frameworks, and explainable and trustworthy reliability evaluation. Finally, emerging demands for the future are outlined.

A. Reliability Evaluation: Key Lessons and Insights

This survey systematically reviews the key factors affecting the reliability of optical networks. It also compares several mainstream evaluation methods. It further outlines survivability technologies and examines trade-offs between reliability and cost and between reliability and energy consumption. Based on this, we summarize the following key lessons and insights:

Evaluating the reliability of optical networks is a complex, interdisciplinary task. This complexity has two main implications. First, a solid mathematical foundation is required, including graph theory, optimization theory, communication theory, algorithmic frameworks such as MC and BN, and the principles of optical communications. Additionally, because optical networks are deployed on land, undersea, and in space, environment-specific engineering knowledge is needed, such as understanding pressure effects and seabed repair models in deep-sea settings. Furthermore, both local reliability and global reliability are important. Local reliability focuses on the reliability of individual components or specific fiber segments, while global reliability concerns the reliability of the entire network. For example, in the case of failures, local reliability primarily considers single-node or single-link faults, whereas global reliability addresses cut-set failures that can disconnect the network. According to our study [43], improving local reliability does not always lead to better global reliability. It may even waste resources. A better strategy is to identify critical bottlenecks and eliminate them with targeted measures. For example, adding WSS redundancy modules inside large ROADM nodes can improve local switching reliability. However, in networks that already support end-to-end protection, the extra benefit is small, while the cost is high. This includes increased CAPEX (chassis, boards, and power) and OPEX (energy and maintenance). In summary, effective evaluation must integrate local and global reliability, aligning local improvements with global reliability and service continuity.

B. Practical Experience and Refinement of Existing Models

Although theoretical research has achieved rich results, current mainstream model-driven methods (such as RBD and FT) face scalability and real-time performance challenges in large-scale networks. RBD and FT models cannot describe dynamic changes over time, so they are usually suitable for early-stage static evaluations. In contrast, MC, BN, and agent-based methods can capture time and dependency relationships. This gives them better adaptability in real deployment and dynamic monitoring scenarios. MC, BN, and agent-based methods are regarded as promising evaluation methods for the future. The following section summarizes the lessons learned from these three methods and discusses their potential development.

1) MC: Current MC in optical networks faces several limitations. These challenges arise because practical optical networks are large, complex, and subject to diverse failure and repair behaviors. First, the MC often suffers from state-space explosion in large-scale optical networks. Second, excessive modeling of non-critical redundancies inflates the state space with limited benefit. Third, the restriction to exponentially distributed failure and repair times fails to capture the variability observed in practice, such as Weibull-distributed lifetimes. Fourth, the assumption of independent failures is frequently invalid, as correlated events such as SRLG fiber cuts occur.

One promising solution to state-space explosion is state-space compression through multi-level dimensionality reduction and graph-based modeling. Large networks can be partitioned into functional subnetworks, with MC applied only to critical nodes and links, while redundant elements are abstracted as equivalent states to simplify computation. To mitigate the impact of non-critical redundancies, importance analysis (e.g., Birnbaum) and cut/path screening can be performed, retaining only high-importance components while compressing the rest into equivalent availability parameters or simplified k-out-of-n structures. To relax the limitation of exponential assumptions, semi-Markov processes (SMPs) allow arbitrary sojourn time distributions while retaining the Markovian property for state transitions, making them well-suited for modeling non-exponential failure and repair times. Finally, to handle dependent failures, explicit modeling of SRLG and common-cause failures or the integration of MC with BN can be applied to capture such dependencies.

This article has been accepted for publication in IEEE Communications Surveys & Tutorials. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/COMST.2025.3621165

2) BN: BN is a useful tool for modeling uncertainty and dependencies between variables. They show good potential in the reliability evaluation of optical networks. However, there are still many challenges in practical use. First, the modeling process, especially building the CPTs, often depends heavily on expert knowledge. This leads to strong subjectivity and low model reproducibility. Second, high-quality labeled data are often missing in real networks. For example, it is hard to obtain accurate mappings between failure states and root causes. As a result, BN faces limitations in real-world engineering applications based on structure learning and parameter learning methods. Third, BN inference has a high computational cost in complex scenarios with many nodes and links. Existing tools such as Netica and GeNIe often cannot meet real-time requirements. Efficient, robust, and deployable approximate inference algorithms for large-scale and dynamic optical network reliability evaluation are still under development. Finally, there is no unified model validation system. Most current studies are based only on simulations or comparison experiments. They lack validation using real network data. This reduces the credibility and practical value of the models.

In the future, BN development can follow several directions. With the growth of industrial IoT and smart sensors, data such as optical power, BER, and alarm logs can be collected in real time. These data can be combined with historical failure records. Based on these data, classification models can be used to determine the status of devices. Then, the component status can serve as prior knowledge in the inference process of the DBN. In this way, the model can perform reliability inference of optical networks when observations are available, which improves both real-time performance and the accuracy of the results. Furthermore, ML techniques can be incorporated to enhance model training, adapt parameter settings to changing network topologies, and improve the efficiency of DBN construction. Finally, approximate inference methods such as particle filtering or Monte Carlo sampling can reduce computing costs while maintaining accuracy. This helps improve the real-time performance of BN in large-optical networks.

3) Agent-Based Reliability Evaluation Framework: Currently, agent-based reliability evaluation frameworks show great potential in adapting to dynamic environments. A multi-level agent framework drives optical network reliability evaluation beyond traditional static RBD and MC toward a new dynamic, autonomous, and interactive paradigm. The collaboration among task management agents, network agents, component agents, and environment agents enables the system to continuously track the full degradation, failure, and repair lifecycle of key components such as fibers, EDFAs, and OXCs in minute-level simulations. While this approach offers high modeling accuracy, it also presents some cost-related challenges. On the one hand, as state spaces increase, the computational burden escalates sharply. Without hardware acceleration, such as graphics processing units (GPUs) or field-programmable gate arrays (FPGAs), achieving real-time execution in large-scale networks is challenging. On the other hand, modeling of failure and repair rates relies heavily on field statistical data. When historical data are insufficient, one may incorporate online measurements or adopt non-parametric estimation methods to reduce bias. Agent-based frameworks are expected to evolve in three key directions.

- Online monitoring: more accurate reliability tracking and early fault warnings can be achieved by synchronizing agent states with real-time telemetry data.
- Cross-domain evaluation: joint modeling and coordinated evaluation can be realized by considering interdependencies between optical networks and other critical infrastructures such as power grids.
- Closed-loop control: maintenance and routing strategies can be dynamically executed based on evaluation results, such as replacing high-risk components or rerouting traffic from potential failure links.

To achieve these objectives, agent-based reliability evaluation frameworks must integrate SDN-based telemetry with programmable optical-layer monitoring. This integration allows real-time collection of performance metrics such as OSNR and BER. In addition, the frameworks should incorporate failure classification and time-series prediction models, enabling a closed loop of reliability evaluation, dynamic control, and energy-efficient scheduling. Future work will focus on three directions: building standardized FIT/MTTR databases, enhancing open-source AnyLogic model repositories, and developing lightweight solvers for efficient execution on GPU or FPGA platforms. These efforts will drive agent-based reliability evaluation automation, from data acquisition and model design to online evaluation and closed-loop control.

C. Future-Oriented Optical Network Reliability Evaluation

1) RAMS-Based Evaluation: Future optical network reliability evaluation must go beyond a single reliability metric and adopt a multi-index comprehensive analysis approach to evaluate overall network performance. In optical networks, RAMS optimization encompasses several key aspects, including reliability (e.g., failure rate and component lifespan), availability (e.g., topology structure and user satisfaction), maintainability (e.g., maintenance efficiency, diagnostic capability, and maintenance procedures), and safety (e.g., resilience against attacks and data protection). A systematic RAMS evaluation and optimization of optical networks enables more precise control and management, facilitating optimal decision-making and enhancing network stability and service quality.

2) AI-Driven Reliability Evaluation: AI can learn from large and heterogeneous telemetry data (such as alarms, BER, and OSNR), and provide probabilistic forecasts with quantified uncertainty for risk-aware decisions. The key enabling techniques for AI-driven reliability evaluation of optical networks include hybrid modeling that integrates physics-based knowledge with machine learning, Bayesian uncertainty quantification, causal reasoning, semi- or self-supervised learning when labeled data are scarce, and online or continual learning to handle distribution drift. Building on this, AI-driven reliability evaluation views the network as a dynamic system that continuously integrates multi-level data from components, links, and topology, enabling real-time fault detection, diagnosis, degradation forecasting, remaining useful life estimation, and risk assessment [368]. Compared with traditional model-based methods, this approach uncovers hidden dependencies, maintains accuracy under non-stationary conditions, and supports adaptive, closed-loop network management even with limited historical data.

3) Digital Twin-Assisted Reliability Evaluation: Digital twin technology is a key enabler for automating optical network reliability evaluation, offering real-time simulation and predictive analysis [369]. Creating a virtual replica of the network integrates topology, fiber characteristics, component parameters, and transmission data, minimizing data limitations and reducing trial-and-error costs [370]. This technology enables real-time monitoring of component status, failure detection, and proactive reliability evaluation, preventing service disruptions [371]. As shown in Fig. 21, The open line system (OLS) controller collects network data from monitoring systems, sensors, and historical records, which is processed through virtualization and a multi-mapping model to maintain an updated view of the network [372]. This digital twin-driven approach ensures reliable network operation, supports network planning and enhances optical network management without interfering with real-world deployments.

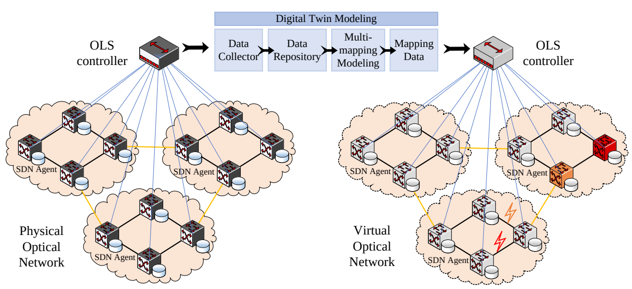

Fig. 21. The evaluation model based on digital twin.

4) Integrated Closed-Loop Reliability Evaluation: Traditional reliability evaluation in optical networks is predominantly open-loop and offline: models (e.g., RBD and MC) are parameterized statically, and operators act on the results manually, with no feedback to recalibrate the models. In contrast, an integrated closed-loop approach fuses continuous telemetry, an online evaluator, and an automated control plane into a sense–evaluate–act–learn cycle. Operational data (alarms, BER/OSNR, flow statistics, and configuration changes) are continuously assimilated to infer reliability states and forecast failure risk in real time [373]. Upon detecting degradation or elevated risk, the controller proactively invokes mitigations (e.g., SDN-based rerouting, transponder parameter tuning, protection switchover, or resource reconfiguration) without human intervention. Outcomes are then fed back to update model parameters and policies, improving subsequent decisions. Compared with open-loop, offline analyses, this closed-loop design reduces MTTR, raises service availability, and sustains accuracy under non-stationary conditions through continual learning.

5) Explainable and Trustworthy Reliability Evaluation: While considering the applicability of a reliability evaluation model, it is also essential to ensure that the evaluation results are explainable and trustworthy. In current research, we found that the explainability and trustworthiness of reliability evaluation results are often overlooked. An explainable reliability evaluation should provide a clear cause-and-effect chain for each decision, identifying the triggers of reliability degradation, such as critical link failures, and quantifying their impact on service quality. Explainability can be ensured through measures such as constructing explicit cause-and-effect chains to trace reliability degradation, visualizing fault propagation paths to highlight critical events, and quantifying the contribution of key parameters using explainable AI techniques, for example, sensitivity analysis and shapley additive explanations (SHAP) [374], thereby making the evaluation process transparent and interpretable. A trustworthy reliability evaluation should ensure the results remain accurate, verifiable, and stable under varying data conditions. Trustworthiness can be ensured through multiple measures, such as using redundant monitoring data to cross-verify key parameters and applying domain knowledge to conduct failure mode and effects analysis to identify risks caused by model bias or missing data. In addition, trustworthy reliability evaluation should include statistical confidence intervals, such as upper and lower bounds derived from MCSs, to quantify uncertainty and guide operational decisions at different confidence levels [375]. For example, a reliability score of 98% with a standard deviation of 1.5% may provide weaker assurance than a score of 97% with a deviation of only 0.2%. Incorporating explainability and trustworthiness into reliability evaluation ensures that the results can be used with assurance for network control, thereby reducing the likelihood of misguided decisions caused by model bias or missing data and improving the resilience of optical networks under dynamic operating conditions.

D. New Requirements in Reliability Evaluation

1) Reliability Evaluation in Interdependent Networks: Modern networks exhibit complex interdependencies compared to past networks that provided independent functionalities [376]. For example, optical networks depend on the electrical grid for power, while the grid relies on optical networks for transmission control signals [377]. Optical networks with active components like lasers and EDFA require a stable power supply to function properly. For instance, a cascade failure between San Diego's power grid and optical networks in 2011 caused a large-scale outage [378]. These examples highlight the need to consider power grid interdependencies in optical network reliability evaluations.

2) SAGIN: With the accelerating demand for ubiquitous high-speed connectivity, the structural limits of terrestrial optical networks in coverage and flexibility have become increasingly evident. Deploying optical transmission lines in extreme terrains such as mountainous, desert, or polar regions presents significant challenges due to harsh environmental conditions, high deployment costs, and complex maintenance requirements, which hinder high-speed and reliable communication in remote areas [379]. The SAGIN addresses these limitations by integrating terrestrial, aerial, and satellite segments to enhance coverage and reliability [380]. However, ensuring the reliability of SAGIN is more complex than in conventional terrestrial networks due to dynamic topology changes, frequent link switching, mobility management, and spectrum allocation challenges [381]. The mobility of satellites requires frequent handovers between satellite–ground and inter-satellite links, resulting in time-varying network topologies and potential connection instability. In addition, factors such as Doppler shifts, delays in channel state information, atmospheric conditions, and ground station placement further influence network reliability [382]. These challenges highlight the need for future research to develop reliability evaluation metrics and dynamic modeling approaches capable of accommodating large-scale SAGIN deployments while considering dynamic topology, mobility, and channel uncertainties.

IX. Conclusion

This survey provides a comprehensive overview of recent advances in the reliability evaluation of optical networks, a topic of growing importance in both academia and industry. It fills an important gap in the literature by systematically distinguishing related concepts such as reliability, availability, and survivability, and by summarizing major research directions, including factors influencing reliability, evaluation metrics and methodologies, and approaches for enhancing network reliability. Trade-offs between reliability and cost and between reliability and energy consumption are also analyzed, offering valuable insights for network design and operation. The main contribution of this survey lies in the systematic examination of reliability evaluation methods. In addition to presenting essential background knowledge, a detailed investigation of representative methods is conducted, key lessons and insights are synthesized, and open challenges are highlighted. These contributions facilitate the operators' design of more reliable networks and support the development of improved evaluation techniques by researchers as technologies evolve. For newcomers to the field, the survey serves as a structured entry point that helps avoid blind exploration and accelerates research engagement.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China (Nos. 62171050, 62125103), Fundamental Research Funds for the Central Universities (2024ZCJH02), and Open Fund of State Key Laboratory of Information Photonics and Optical Communications (BUPT, No. IPOC2021ZT15).

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